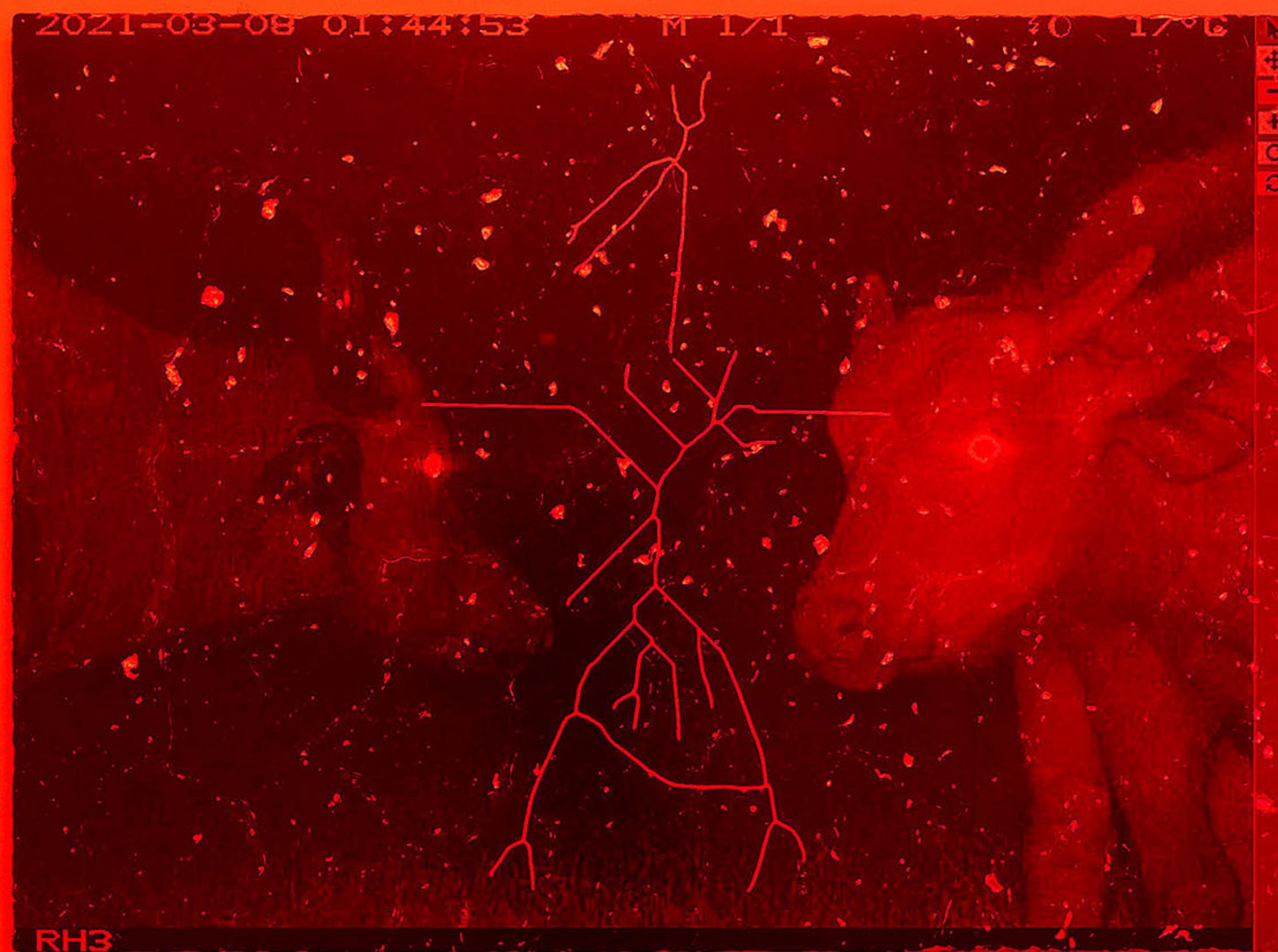


On Aesthetics



**Where
We Have
Been
and
Where
We Are
Going**



We are surrounded by a new visual language, one born from the collision of the digital and the physical. This publication is a guide to this unfamiliar terrain. It begins by deciphering the origins of our post-digital reality, where the aesthetics of the internet seeped into our everyday world. Now, we venture further into the frontier of a world generated by AI, a realm of synthetic imagery and algorithmic creation. Here, we confront a pressing question: as machines learn our aesthetics, are we witnessing a renaissance or a great homogenization? This is an exploration of the strange beauty and profound challenges of art in the age of artificial creation.

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Internet Cache
Self Portrait
Eva Roth (2012- 2014)

A Search for
New Concepts

Understanding
Post- Digital

What is
Post- Internet?

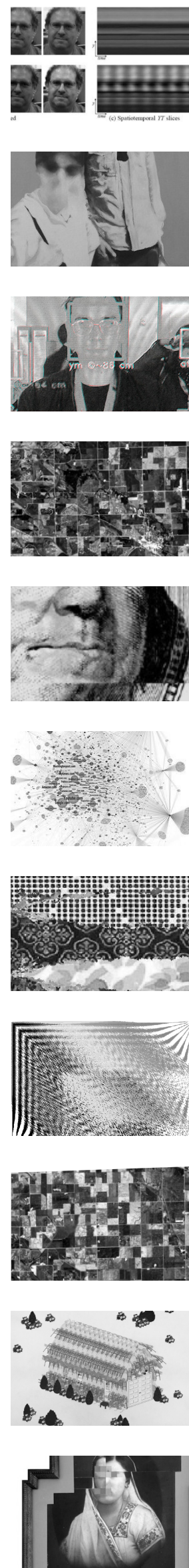
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Thomas Ruff (2007)

The Web
Spilled Over

The New
Aesthetic
James Bridle (20011)

Everything about
“The New Aesthetic”

Where We Have Been



“But if digital technologies become more and more efficient in ‘perceiving’ the real world, what does this mean concerning our notion of aesthetic, a ‘perception from the senses?’”

A Search for New Concepts

“Post-internet, postdigital, New Aesthetic”

Thinking Postdigital Aesthetics:
Art, Computation and Design.
David M. Berry and Michael Dieter, 2015



The everyday experience of life within computational societies inspires a search for new concepts and experiences, or perhaps ‘formal indicators’ as vague neologisms, in an attempt to historically delimit and define the present. Accordingly, in different ways, notions such as post-internet, postdigital and the new aesthetic can be taken as attempts to grapple with the immersive and disorientating experiences of computational infrastructures as they scale up and intensify.

Indeed, the revival of Félix Guattari’s concept of post-media can additionally be understood within this context of a search for orienting alternatives to counter the current trajectories of digitalization (Apprigh et al. 2013; Quaranta 2013). It would be inaccurate, or at least too easy, to quickly dismiss these terms as simply offline Romanticism, art world jargon or stylized cases of hipster analogue culture. On the contrary, we argue that they can be read as connected instances of an effort to collectively develop concepts that reflect on the non-neotic aspect of the digital. That is, as ubiquitous computational infrastructures radiate data, they encourage tacit modes of knowing and the iteration of habit— and thus also create agnosis, or ‘not knowing’, through a form of agnotology. By ‘agnotology’ we are referring to the way in which computation facilitates a systemic production and maintenance of ignorance. Computational technologies direct us towards a passive trust in widely delegated, yet obfuscated, actions (see Berry 2012). This tendency towards automated and accelerated modes of action complicates and may undermine structures of reflection and critique. One consequence is a twisting and turning of computational logics into other contexts against attempts to orient and ‘get a grip’ on computational things. In this way, notions such as the postdigital are also performed and mediatized in rather novel ways, and can be taken as a complementary unfolding of an aesthetization of computational infrastructures.

This can be seen in the emergence of the New Aesthetic as a project initiated by James Bridle, a British designer and programmer, as an attempt to document and catalogue patterns of the computational throughout every daylife. The NewAesthetic, therefore, signals a kind of threshold or saturation point whereby the obscure ubiquity of digital, networked and mobile devices inspires a struggle to map, document and record; in other words, to make sensible and intelligible the seemingly opaque operations of digital infras tructure, even while invoking an ambiguous gesture of aesthetization using the Tumblr.com platform. Working from within an explicitly art-world context, the notion of post-internet art has, meanwhile, been elaborated as works that engage with digital networking through hybrid, often offline, manifestations. In this sense, artist and curator Marisa Olson uses the term to describe art literally created after internet use: the creative ‘yield’ from hours of consumptive downloading and browsing (Debatty 2008). This can, more over, be taken as a situation of art making after the internet has massified through platformization, resulting in a mainstreaming marked by the shift from exceptional to ordinary perceptions of digital creativity.

The role of the practitioner here, then, is also imagined in terms of techniques of recognizing patterns, cataloguing, curating, interpreting and transcribing (Vierkant 2010), and then actualizing these engagements as artefacts for potential contemplation. The ‘postdigital’, meanwhile, also covers a wide range of issues attached to the entanglements of media life after the digital, including a shift from an earlier moment driven by an almost obsessive fascination and enthusiasm with new media to a broader set of affectations that now includes unease, fatigue, boredom and disillusionment. Linked to ideas like the ‘off-internet’ and ‘neo-analogue’, the postdigital recognizes the revival of ‘old’ media formats like cassette tapes or analogue synthesizers, and more generally maps out ‘the messy state of media, arts and design after their digitization’ (Cramer 2015, this volume). Crucially, this also involves working through the implementations of the computal in a regular state of constant upheaval. In other words, this is a condition in which digital disruption is not transcended as such, but becomes routine or business as usual.

All of these proposed terms and concepts seize on a hybridized approach towards the digital and non-digital, finding characteristics of one within the other, deliberately mixing up processes of making things discrete, calculable, indexed and automated in unorthodox ways. In doing so, they form part of an epistemological asterism of practices, experiences and mediations that follows the primacy of the computal as normative. That is, the appearance of these terms can be interpreted collectively as endeavours to elucidate the trajectories of ubiquitous digitalization; they collectively form new patterns which can help us begin to map and historicize the varieties of computal societies.

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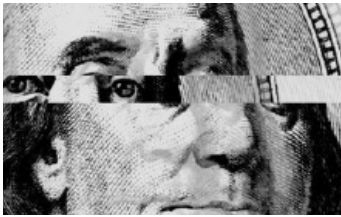
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Eva Roth (2012- 2014)

[A] Roth, E. (n.d.). Internet Cache Self Portrait series.
Retrieved 12/11/2025

Internet Cache Self Portrait

The Internet Cache Self Portrait series are composed of uncensored streams of images passively collected through daily Internet browsing. Faces of “friends” from social media exist side by side with corporate logos, mangled pieces of google maps, family photos and banner advertisements. These algorithmically produced prints act as a contemporary nude, exposing in a generous and open way the subjects’s private online interactions.

The Internet Cache Self Portrait series attempts to reveal something human and intimate about us through the lost narratives left behind in our interactions online. While the work puts the artist as the focus, it is also painting a portrait of the Internet at this specific moment in time. The prints, composed of memories that were never intended to be saved, are archiving the seemingly incidental into something permanent. [A]



Understanding Post-Digital

“What Is ‘Post-digital’?”

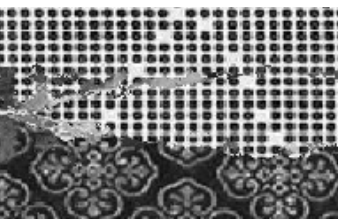
Thinking Postdigital Aesthetics: Art,
Computation and Design. Florian Cramer,
2015

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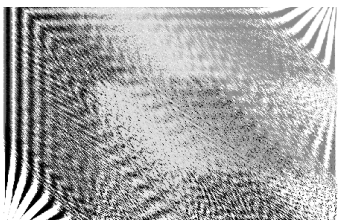
The meme, which was still making news at the time of writing this paper in late 2013 (Hermelin 2013), nicely illustrates the rift between ‘digital’ and ‘post-digital’ cultures. Imageboard memes are arguably the best example of a contemporary popular mass culture which emerged and developed entirely on the internet. Unlike earlier popular forms of visual culture such as comic strips, they are anonymous creations— and, as such, even gave birth to the now-famous Anonymous movement, as described by Klok (2010, 16–19). In January 2013, a picture of a young man typing on a mechanical typewriter while sitting on a park bench went ‘viral’ on the popular website Reddit. The image was presented in the typical style of an ‘image macro’ or ‘imageboard meme’ (Klok 2010, 16–19), with a sarcastic caption in bold white Impact typeface that read: ‘You’re not a real hipster— until you take your typewriter to the park.’ Other important characteristics of imageboard memes are: creation by users, disregard of intellectual property, viral dissemination among users, and potentially infinite repurposing and variation (through collage or by changing the text). As low-resolution images with



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small file sizes, they can be created and disseminated almost instantly, in contrast with the much slower creation, editing and distribution processes characteristic of traditional publishing media. The 'digital' imageboard meme portrays the 'analogue' typewriter hipster as its own polar opposite—in a strictly technical sense, however, even a mechanical typewriter is a digital writing system, as I will explain later in this chapter. Also, the typewriter's keyboard makes it a direct precursor of today's personal computer systems, which were used for typing the text of the imageboard meme in question. Yet, in a colloquial sense, the typewriter is definitely an 'analogue' machine, as it does not contain any computational electronics. In 2013, using a mechanical typewriter rather than a mobile computing device is, as the imageboard meme suggests, no longer a sign of being old-fashioned. It is, instead, a deliberate choice of renouncing electronic technology, thereby calling into question the common assumption that computers, as meta-machines, represent obvious technological progress and therefore constitute a logical upgrade from any older media technology much in the same way as using a bike today calls into question the assumption, common in many Western countries since World War II, that the automobile is by definition a rationally superior means of transportation, regardless of the purpose or context.

Typewriters are not the only medium which has recently been resurrected as literally a post-digital device: other examples include vinyl records, and more recently also audio cassettes, as well as analogue photography and artists' printmaking. And if one examines the work of contemporary young artists and designers, including art school students, it is obvious that these 'old' media are vastly more popular than, say, making imageboard memes.

DISENCHANTMENT WITH 'DIGITAL'

I was first introduced to the term 'post-digital' in 2007 by my then-student Marc Chia— now Tara Transitory, also performing under the moniker One Man Nation. My first reflex was to dismiss the whole concept as irrelevant in an age of cultural, social and economic upheavals driven to a large extent by computational digital technology. Today, in the age of ubiquitous mobile devices, drone wars and the gargantuan data operations of the US National Security Agency (NSA), Google and other global players, the term may seem even more questionable than it did in 2007: as either a sign of ignorance of our contemporary reality, or else of some deliberate Thoreauvian—Luddite withdrawal from this reality (see also Berry 2015, this volume). More pragmatically, the term 'post-digital' can be used to describe either a contemporary disenchantment

with digital information systems and media gadgets, or a period in which our fascination with these systems and gadgets has become historical— just as the dot-com age ultimately became historical in the 2013 novels of Thomas Pynchon and Dave Eggers. After Edward Snowden's disclosures of the NSA's all-pervasive digital surveillance systems, this disenchantment has quickly grown from a niche 'hipster' phenomenon to a mainstream position—one which is likely to have a serious impact on all cultural and business practices based on networked electronic devices and internet services.ww

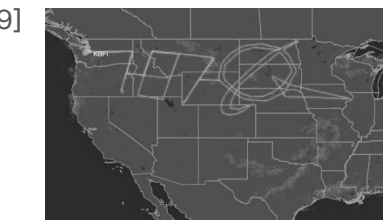
REVIVAL OF 'OLD MEDIA'

While a Thoreauvian—Luddite digital withdrawal may seem a tempting option for many, it is fundamentally a naïve position, particularly in an age when even the availability of natural resources depends on global computational logistics, and intelligence agencies such as the NSA intercept paper mail as well as digital communications. In the context of the arts, such a withdrawal seems little more than a rerun of the 19th-century Arts and Crafts movement, with its programme of handmade production as a means of resistance to encroaching industrialization. Such (romanticist) attitudes undeniably play an important role in today's renaissance of artists' print making, handmade film labs, limited vinyl editions, the rebirth of the audio cassette, mechanical typewriters, analogue cameras and analogue synthesizers. An empirical study conducted by our research centre Creating 010 in Rotterdam among Bachelor students from most of the art schools in the Netherlands indicated that contemporary young artists and designers clearly prefer working with non-electronic media: given the choice, some 70% of them 'would rather design a poster than a website' (Van Meer 2012, 2013, 14). In the Netherlands at least, education programmes for digital communication design have almost completely shifted from art academies to engineering schools, while digital media are often dismissed as commercial and mainstream by art students (Van Meer 2012–2013, 5). Should we in turn dismiss their position as romanticist and neo-Luddite?

POST- DIGITAL = POSTCOLONIAL POST- DIGITAL = POST- HISTOIRE

On closer inspection, however, the dichotomy between digital big data and neo-analogue do-it-yourself (DIY) is really not so clear-cut. Accordingly, 'post-digital' is arguably more than just a sloppy descriptor for a contemporary (and possibly nostalgic) cultural trend. It is an objective fact that the age in which we now live is not a post-digital age, either in terms of technological developments— with

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no end in sight to the trend towards further digitization and computerization— or from a historico-philosophical perspective. Regarding the latter (Cox 2015, this volume) offers a valid critique of the ‘periodizing logic’ embedded in the term ‘post-digital’, which places it in the dubious company of other historico-philosophical ‘post’-isms, from postmodernism to post-histoire. However, ‘post-digital’ can be defined more pragmatically and meaningfully within popular cultural and colloquial frames of reference. This applies to the prefix ‘post’ as well as the notion of ‘digital’. The prefix ‘post’ should not be understood here in the same sense as postmodernism and post-histoire, but, rather, in the sense of post-punk (a continuation of punk culture in ways which are somehow still punk, yet also beyond punk); post-communism (as the ongoing social–political reality in former Eastern Bloc countries); post-feminism (as a critically revised continuation of feminism, with blurry boundaries with ‘traditional’, unprefixes feminism); postcolonialism (see next paragraph); and, to a lesser extent, post apocalyptic (a world in which the apocalypse is not over, but has progressed from a discrete breaking point to an ongoing condition— in Heideggerian terms, from Ereignis to Being— and with a contemporary-popular iconography pioneered by the Mad Max films in the 1980s).

None of these terms— post-punk, post-communism, post-feminism, postcolonialism, post-apocalyptic— can be understood in a purely Hegelian sense of an inevitable linear progression of cultural and intellectual history. Rather, they describe more subtle cultural shifts and ongoing mutations. Postcolonialism does not in any way mean an end of colonialism (akin to Hegel’s and Fukuyama’s ‘end of history’), but, rather, its mutation into new power structures, less obvious but no less pervasive, which have a profound and lasting impact on languages and cultures, and, most significantly, continue to govern geopolitics and global production chains. In this sense, the post-digital condition is a post-apocalyptic one: the state of affairs after the initial upheaval caused by the computerization and global digital networking of communication, technical infrastructures, markets and geopolitics..

DIGITAL = STERILE HIGH TECH?

Also, the ‘digital’ in ‘post-digital’ should not be understood in any technical scientific or media-theoretical sense, but, rather, in the way the term is broadly used in popular culture— the kind of connotation best illustrated by a recent Google Image Search result for the word ‘digital’. The first thing we notice is how the term ‘digital’ is, still in 2013, visually associated with the colour blue. Blue is literally the coolest colour in the colour spectrum (with a temperature

of 15,000–27,000 Kelvin), with further suggestions of cultural coolness and cleanness. The simplest definition of ‘post-digital’ describes a media aesthetics which opposes such digital high-tech and high-fidelity cleanness. The term was coined in 2000 by the musician Kim Cascone, in the context of glitch aesthetics in contemporary electronic music (Cascone 2000, 12). Also in 2000, the Australian sound and media artist Ian Andrews used the term more broadly as part of a concept of ‘post-digital aesthetics’, which rejected the ‘idea of digital progress’ as well as ‘a teleological movement toward “perfect” representation’ (Andrews 2000). Cascone and Andrews considered the notion of ‘post-digital’ primarily as an antidote to techno-Hegelianism. The underlying context for both their papers was a culture of audio-visual production in which ‘digital’ had long been synonymous with ‘progress’: the launch of the Fairlight CMI audio sampler in 1979, the digital audio CD and the MIDI standard (both in 1982), software-only digital audio workstations in the early 1990s, real-time programmable software synthesis with Max/MSP in 1997.

Such teleologies are still prevalent in video and TV technology, with the ongoing transitions from SD to HD and 4K, from DVD to BluRay, from 2D to 3D— always marketed with a similar narrative of innovation, Ironically, the use of the term ‘post-digital’ was somewhat confusing in the context of Cascone’s paper, since the glitch music defined and advocated here actually was digital, and even based on specifically digital sound processing artefacts. On the other hand, and in the same sense as post-punk can be seen as a reaction to punk, Cascone’s concept of ‘post-digital’ may best be understood as a reaction to an age in which even camera tripods are being labelled as ‘digital’, in an effort to market them as new and superior technology improvement and higher fidelity of reproduction. In rejecting this narrative, Cascone and Andrews opposed the paradigm of technical quality altogether.

DIGITAL = LOW QUALITY TRASH?

There is a peculiar overlap between, on the one hand, a post-digital rejection of digital high tech and, on the other hand, a post-digital rejection of digital low quality. Consider, for example, the persisting argument that vinyl LPs sound better than CDs (let alone MP3s); that film photography looks better than digital photography (let alone smartphone snapshots); that 35 mm film projection looks better than digital cinema projection (let alone Bit-Torrent video downloads or YouTube); that paper books are a richer medium than websites and e-books; and that something typed on a mechanical typewriter has more value than a throwaway digital text file (let alone e-mail

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Images gathered from “The New Aesthetic” Tumblr Archive



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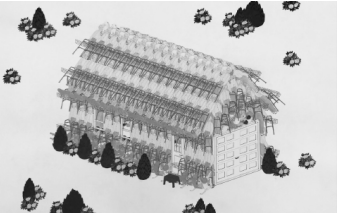


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Images gathered from “The New Aesthetic” Tumblr Archive

spam). In fact, the glitch aesthetics advocated by Cascone as ‘post-digital’ are precisely the same kind of digital trash dismissed by ‘post-digital’ vinyl listeners.

(The following is an attempt to recapitulate and order some observations which I have formulated in previous publications.)

POST- DIGITAL = POST- DIGITIZATION

Returning to Cascone and Andrews, but also to post-punk, postcolonialism and Mad Max, the term ‘post-digital’ in its simplest sense describes the messy state of media, arts and design after their digitization (or, at least, the digitization of crucial aspects of the channels through which they are communicated). Sentiments of disenchantment and scepticism may also be part of the equation, though this need not necessarily be the case sometimes ‘post-digital’ can, in fact, mean the exact opposite.

Contemporary visual art, for example, is only slowly starting to accept practitioners of net art as regular contemporary artists— and then again, preferably those like Cory Arcangel, whose work is white cube-compatible. Yet its discourse and networking practices have been profoundly transformed by digital media such as the e-flux mailing list, art blogs and the electronic e-flux journal. In terms of circulation, power and influence, these media have largely superseded printed art periodicals, at least as far as the art system’s in-crowd of artists and curators is concerned. Likewise, when printed newspapers shift their emphasis from daily news (which can be found more quickly and cheaply on the internet) to investigative journalism and commentary— like The Guardian’s coverage of the NSA’s PRISM programme they effectively transform themselves into post-digital or post-digitization media.

POST- DIGITAL = ANTI- ‘NEW MEDIA’

‘Post-digital’ thus refers to a state in which the disruption brought upon by digital information technology has already occurred. This can mean, as it did for Cascone, that this technology is no longer perceived as disruptive. Consequently, ‘post-digital’ stands in direct opposition to the very notion of ‘new media’. At the same time, as its negative mirror image, it exposes arguably even deconstructs— the latter’s hidden teleology: when the term ‘post-digital’ draws critical reactions focusing on the du-

bious historico philosophical connotations of the prefix ‘post’, one cannot help but wonder about a previous lack of such critical thinking regarding the older (yet no less Hegelian) term ‘new media’.

POST- DIGITAL = HYBRIDS OF ‘OLD’ AND ‘NEW MEDIA’

At the same time, new ethical and cultural conventions which became mainstream with internet communities and Open Source culture are being retroactively applied to the making of non-digital and post-digital media products. A good example of this is collaborative zine conventions, a thriving subculture documented on the blog fanzines. tumblr.com and else where. These events, where people come together to collectively create and exchange zines (i.e. small-circulation, self-published magazines, usually focusing on the maker’s cultural and/or political areas of interest), are infact the exact opposite of the ‘golden age’ zine cultures of the post-punk 1980s and 1990s, when most zines were the hyper-individualistic product and personality platforms of one single maker. If we were to describe a contemporary zine fair or mimeography community art space using Lev Manovich’s new media taxonomy of ‘Numerical Representation’, ‘Modularity’, ‘Automation’, ‘Variability’ and ‘Transcoding’ (Manovich 2002b,27–48), then ‘Modularity’, ‘Variability’ and— in a more loosely metaphorical sense— ‘Transcoding’ would still apply to the contemporary cultures working with these ‘old’ media. In these cases, the term ‘post-digital’ usefully describes ‘new media’-cultural approaches to working with so-called ‘old media’.

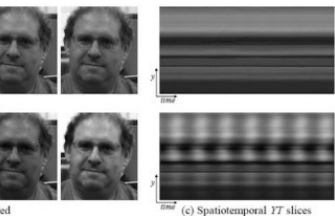
DO- IT- YOURSELF VS. CORPORATE MEDIA RATHER THAN ‘NEW’ VS. ‘OLD’ MEDIA

When hacker-style and community-centric working methods are no longer specific to ‘digital’ culture (since they are now just as likely to be found at an ‘analogue’ zine fair as in a ‘digital’ computer lab), then the established dichotomy of ‘old’ and ‘new’ media— as synonymous in practice with ‘analogue’ and ‘digital’— becomes obsolete, making way for a new differentiation: one between shrink-wrapped culture and do-it-yourself culture.

The best example of this development (at least among mainstream media) is surely the magazine and website Make, published by O’Reilly since 2005, and instrumental for the foundation of the contemporary ‘maker movement’.



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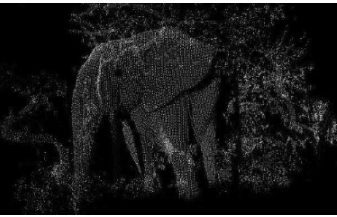
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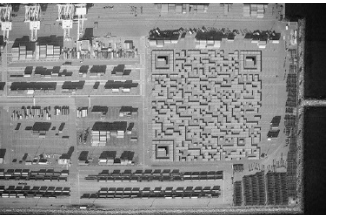
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Images gathered from “The New Aesthetic” Tumblr Archive

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Images gathered from “The New Aesthetic” Tumblr Archive



[21]



Make covers 3D printing, Arduino hardware hacking, fab lab technology, as well as classical DIY and crafts, and hybrids between various ‘new’ and ‘old’ technologies. The 1990s/early 2000s assumption that ‘old’ mass media such as newspapers, movies, television and radio are corporate, while ‘new media’ such as websites are DIY, is no longer true now that user-generated content has been co-opted into corporate social media and mobile apps. The internet as a self run alternative space— central to many online activist and artist projects, from The Thing onwards— is no longer taken for granted by anyone born after 1990: for younger generations, the internet is associated mainly with corporate, registration-only services.

SEMIOTIC SHIFT TO THE INDEXICAL

The ‘maker movement’— as manifested in fab labs, but also at zine fairs represents a shift from the symbolic, as the preferred semiotic mode of digital systems (and of which the login is the perfect example), towards the indexical: from code to traces, and from text to context. 1980s post-punk zines, for example, resembled the art manifestos of the 1920s Berlin Dadaists, while 1980s Super 8 films, made in the context of the Cinema of Transgression and other post-punk movements, proposed underground narratives as an alternative to mainstream cinema. The majority of today’s zines and experimental Super 8 films, however, tend to focus less on content and more on pure materiality, so that the medium, such as paper or celluloid, is indeed the message— a shift from semantics to pragmatics, and from metaphysics to ontology.

TECHNICALLY, THERE IS NO SUCH THING
AS ‘DIGITAL- MEDIA’ OR ‘DIGITAL
AESTHETICS’

Media, in the technical sense of storage, transmission, computation and display devices, are always analogue. The electricity in a computer chip is analogue, as its voltage can have arbitrary, undifferentiated values within a specific range, just like a fretless violin string. Only through filtering can one make a certain sub-range of high voltages correspond to a ‘zero’ and another sub-range of low voltages to a ‘one’. Hardware defects can cause bits to flip, turning zeroes into ones and vice-versa. Also, the sound waves produced by a sound card and a speaker are analogue, for example. This is what Kittler (1992, 81–90) refers to, somewhat opaquely, when he argues that in computing ‘there is no software’. An LCD screen is a hybrid digital analogue system: its display is made of discrete, countable, single pixels, but the light emitted by these pixels can be measured on an analogue continuum. Consequently, there is no such thing as digital media,

only digital or digitized information: chopped-up numbers, letters, symbols and any other abstracted units, as opposed to continuous, wave-like signals such as physical sounds and visible light. Most ‘digital media’ devices are in fact analogue-to-digital-to-analogue converters: an MP3 player with a touch screen interface, for example, takes analogue, non-discrete gesture input and translates it into binary control instructions, which in turn trigger the computational information processing of a digital file, ultimately decoding it into an analogue electrical signal, which another analogue device, the electromagnetic mechanism of a speaker or headphone, turns into analogue sound waves. The same principle applies to almost any so-called digital media device, from a photo or video camera to an unmanned military drone. Our senses can only perceive information in the form of non discrete signals such as sound or light waves. Therefore, anything aesthetic (in the literal sense of aisthesis, perception) is, by strict technical definition, analogue.

DIGITAL = ANALOGUE = POST- DIGITAL...?

A ‘digital artwork’ based on the strictly technical definition of ‘digital’ would most likely be considered ‘post-digital’ or even ‘retro analogue’ by art curators and humanities scholars: for example, stone mosaic floors made from internet imageboard memes, mechanical typewriter installations, 6 count downloops running in Super 8 or 16 mm filmprojection, but also computer installations exposing the indexicality of electrical currents running through circuits. The everyday colloquial definition of ‘digital’ embraces the fiction (or, rather, the abstraction) of the disembodied nature of digital information processing. The colloquial use of ‘digital’ also tends to be metonymical, so that anything connected literally or figuratively to computational electronic devices— even a camera tripod— can nowadays be called ‘digital’. This notion, mainly cultivated by product marketing and advertising, has been unquestioningly adopted by the ‘digital humanities’ (as illustrated by the very term ‘digital humanities’). On the other hand, ‘post-digital’ art, design and media— whether or not they should technically be considered post-digital— challenge such uncritical notions of digitality, thus making up for what often amounts to a lack of scrutiny among ‘digital media’ critics and scholars.

REVISITING THE TYPEWRITER HIPSTER MEME

The alleged typewriter hipster later turned out to be a writer who earned his livelihood by selling custom-written stories from a bench in the park The imageboard meme photo was taken from an angle that left out his sign, taped

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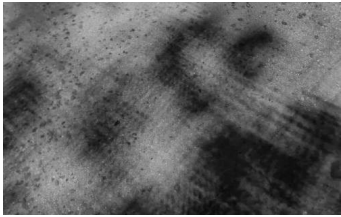


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from
“The New
Aesthetic”
Tumblr
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to his typewriter case: ‘One-of-a-kind, unique stories while you wait’. In an article for the website The Awl, he recollects how the meme made him ‘An Object Of Internet Ridicule’ and even open hatred. Knowing the whole story, one can only conclude that his decision to bring a mechanical type writer to the park was pragmatically the best option. Electronic equipment (a laptop with a printer) would have been cumbersome to set up, dependent on limited battery power, and prone to weather damage and theft, while handwriting would have been too slow, insufficiently legible, and lacking the appearance of a professional writer’s work. not they should technically be considered post-digital– challenge such uncritical notions of digitality, thus making up for what often amounts to a lack of scrutiny among ‘digital media’ critics and scholars. Had he been an art student, even in a media arts programme, the typewriter would still have been the right choice for this project. This is a perfect example of a post-digital choice: using the technology most suit able to the job, rather than automatically ‘defaulting’ to the latest ‘new media’ device. It also illustrates the post-digital hybridity of ‘old’ and ‘new’ media, since the writer advertises (again, on the sign on his typewriter case) his Twitter account ‘@rovingtypist’, and conversely uses this account to promote his story-writing service. He has effectively repurposed the type writer from a prepress tool to a personalized small press, thus giving the ‘old’ technology a new function usually associated with ‘new media’, by exploiting specific qualities of the ‘old’ which make up for the limitations of the ‘new’. Meanwhile, he also applies a ‘new media’ sensibility to his use of ‘old media’: user-customized products, created in a social environment, with a ‘donate what you can’ payment model. Or, rather, the dichotomy of community media vs. mass media has been flipped upside-down, so that a typewriter is now a community media device, while participatory websites have turned into the likes of Reddit, assuming the role of yellow press mass media– including mob hatred incited by wilful misrepresentation.

THE DESIRE FOR AGENCY

Cascone and Andrews partly contradicted themselves when they defined the concept of ‘post-digital’ in the year 2000 Though they rejected the advocacy of ‘new media’, they also heavily relied on it. Cascone’s paper drew on Nicholas Negroponte’s Wired article ‘Beyond Digital’ (Negroponte 1998), while Ian Andrews’ paper referenced Lev Manovich’s ‘Generation Flash’, an article which promoted the very opposite of the analogue/digital, retro/contemporary hybridizations currently associated with the term ‘post digital’ (Manovich 2002a). We could metaphorically describe post-digital cultures as postcolonial practices in a communications world taken over by

a military–industrial complex made up of only a handful of global players. More simply, we could describe these cultures as a rejection of such dystopian techno-utopias as Ray Kurzweil’s and Google’s Singularity University, the Quantified Self movement and sensor-controlled ‘Smart-Cities’. And yet, post-digital subculture, whether in Detroit, Rotterdam or else where, is on a fundamental level not so different from such mainstream Silicon Valley utopias. For Van Meer (2012–2013), the main reason why art students prefer designing posters to designing websites is due to a fiction of agency– in this case, an illusion of more control over the medium. Likewise, ‘digital’ cultures are driven by similar illusions of free will and individual empowerment. The Quantified Self movement, for example, is based on a fiction of agency over one’s own body. The entire concept of DIY, whether non-digital, digital or post-digital, is based on the fiction of agency implied by the very notion of the self-made.

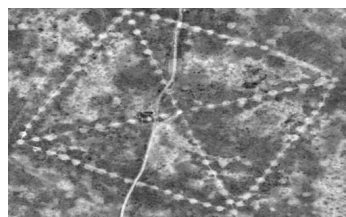
Each of these fictions of agency represents one extreme in how individuals relate to the techno-political and economic realities of our time: either over identification with systems, or rejection of these same systems. Each of these extremes is, in its own way, symptomatic of a systems crisis–notacrisisof this or that system, but, rather, a crisis of the very paradigm of ‘system’, as defined by General Systems Theory, itself an offshoot of cybernetics. A term such as ‘post-Snowden’ describes only one (important) aspect of a bigger picture:8 a crisis of the cybernetic notion of ‘system’ which neither ‘digital’ nor ‘post-digital’– two terms ultimately rooted in systems theory– is able to leave behind, or even adequately describe.

[27] [28]
Images gathered from “The New Aesthetic” Tumblr Archive

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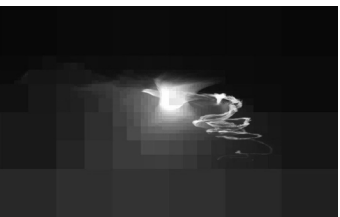
What is Post- Internet?

“Situating Post Internet”

Quaranta, D. “Situating Post Internet”. (II) (III) In
V. Catricalà (Ed.), Media Art. Toward a New
Definition of Art in the Age of Technology,
2015



As is now well known, the term “postinternet” was first used by artist and (at the time) Rhizome editor Marisa Olson in 2006, in reference to her own practice and that of her peers. While she was mainly speaking for herself, as she recently clarified, her post at Rhizome and her role as a cofounder of Nasty Nets, the first surfing club, put her in a position to speak, more broadly, about a new generation of artists who, while spending a lot of time online, were developing most of their work offline: work that was nonetheless “infused with the digital visual language, network aesthetics, and the social politics of online transmission and reception.”



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Images
gathered
from
“The New
Aesthetic”
Tumblr
Archive

The term survived, together with a few alternatives such as internet-engaged art and internet-aware art (the latter first used by artist Guthrie Lonerger) until it was popularized by art critic Gene McHugh, who – thanks to a grant from the Creative Capital | Andy Warhol Foundation Arts Writers Grant Program – ran a blog of the same name from December 2009 to September 2010. [8] While Olson, in her use of the term, stressed the “practical” shift from making art online to making art that took advantage of the act of navigation, McHugh focused instead on the historical and conceptual reasons for this shift. He attributed them to the general transformation of the relationship between reality and the internet, which provided different terrain for the relationship between art and the internet. According to McHugh: As is now well known, the term “postinternet” was first used by artist and (at the time) Rhizome editor Marisa Olson in 2006, in reference to her own practice and that of her peers. While she was mainly speaking for herself, as she recently clarified, her post at Rhizome and her role as a cofounder of Nasty Nets, the first surfing club, put her in a position to speak, more broadly, about a new generation of artists who, “No matter what your deal was/is as an artist, you had/ have to deal with the Internet – not while spending a lot of time online, were developing most of their work offline: work that was nonetheless “infused with the digital visual language, network aesthetics, and the social politics of online transmission and reception.” other successful essays and publications, from Artie Vierkant’s “The Image Object Post-Internet” to Louis Doulas’ “Within Post-Internet, Part One” to Katja Novitskova’s Post Internet Survival Guide and the recent book *You Are Here. Art After the Internet*, edited by Omar Kholeif. The success of the label, of course, went hand-in-hand with the institutional and commercial success of the artists associated with it, and – for the first time in the history of internet engaged art – thanks to the concerted dedication and support of a group of commercial and semi commercial galleries, mostly based in New York, Berlin and London but also further afield, and with a strong online presence.

McHugh’s definition is also important because it links up with the most recent definition of the term, which can be summed up in the words of Karen Archey and Robin Peckham: This understanding of the post-internet refers not to a time “after” the internet, but rather to an internet state of mind – to think in the fashion of the network. In the context of artistic practice, the category of the post-internet describes an art object created with a consciousness of the networks within which it exists, from conception and production to dissemination and reception. But before moving on, it is worth spending some time on the three competing labels that emerged around 2006 – 2008 in the debate around net art. Though very different from one an-

other, post internet, internet-aware and internet-engaged all resulted from an aversion to medium-based labels such as net art or internet art; they all pointed to the internet as a cultural reference, and an environment, rather than a medium. This is related, on the one hand, to a general change in the perception of the relationship between reality and the internet, as McHugh pointed out; and, on the other hand, to a dissatisfaction with medium specificity and the niche status of new media art. After the dotcom bubble and with the arrival of the Web 2.0, the internet started to be perceived less as a medium and more as a key part of our daily lives; less as a utopia to construct together, and more as a dystopia we are all part of, but that still provides interesting opportunities for networking and community making, and an unprecedented tool for “surfing” reality and getting a better understanding of it.

This goes beyond the scope of this essay, but it is worth noting that the idea that the information society has entered a new phase has now been universally accepted, and different people have attempted to describe this shift in different terms, but in very similar ways. More specifically, the debate around post internet art has arisen more or less in parallel to the debates around postmedia, a term first used by Felix Guattari in 1996, post-digital and the New Aesthetic. First used by Kim Cascone in an essay on digital music published in 2000, the notion of post-digital attracted much debate between 2008 and 2014. In 2014, Aarhus University and transmediale, Berlin set up “Post-digital Research”, an initiative that produced a peer-reviewed journal and came up with the following working definition:

Post-digital, once understood as a critical reflection of “digital” aesthetic immaterialism, now describes the messy and paradoxical condition of art and media after digital technology revolutions. “Post-digital” neither recognizes the distinction between “old” and “new” media, nor ideological affirmation of the one or the other. It merges “old” and “new”, often applying network cultural experimentation to analog technologies which it re-investigates and re-uses. It tends to focus on the experiential rather than the conceptual. It looks for DIY agency outside totalitarian innovation ideology, and for networking off big data capitalism. At the same time, it already has become commercialized. The New Aesthetic started as a visual study run on a Tumblr blog by British artist and designer James Bridle in 2011, and gathered momentum around 2012, when author Bruce Sterling made it the subject of an enthusiastic essay in *Wired*. The New Aesthetic blog collects images from art, design, online interfaces and daily life, with a focus on how the digital manifests itself in the physical domain. As Bridle explains in an essay tellingly re-published at the beginning of the book.

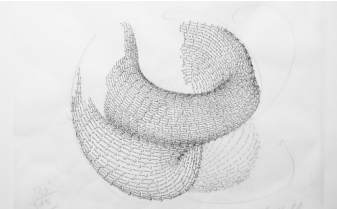
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Images
gathered
from
“The New
Aesthetic”
Tumblr
Archive

It is impossible for me [...] not to look at these images and immediately start to think about not what they look like, but how they came to be and what they become: the processes of capture, storage, and distribution; the actions of filters, codecs, algorithms, processes, databases, and transfer protocols; the weight of datacenters, servers, satellites, cables, routers, switches, modems, infrastructures physical and virtual; and the biases and articulations of disposition and intent encoded in all of these things, and our comprehension of them. Post internet emerged along, and with a deep awareness of, these lines of thought. But the urge to reframe the art formerly known as net art was also related to another process, within the field of new media art: the need to go “beyond new media art”, both as a medium-based practice and as a cultural niche, in order to develop a better dialogue with the contemporary art world after the nonstarters at the turn of the millennium, when new media art missed the chance (offered by a number of museum exhibitions) to be viewed as one of the most interesting artistic developments of the last few years. Around 2005, many “new media artists” - led by representatives of the first net art generation – had already started working with commercial galleries and contemporary art institutions, and investigating ways to present their works in the white cube. Most of them, Jodi included, had all but stopped making net-based projects, and were working mostly on software and hardware manipulation. Two seminal exhibitions, “The Art Formerly Known as New Media” and “Postmedia Condition”, took place that year, the latter introducing the concept of postmedia in the media art debate; a media art festival catalog hosted an equally seminal essay, titled “ ‘It’s contemporary art, stupid’ . Curating computer-based art out of the ghetto”, by curators Inke Arns and Jakob Lillemose.

Later in 2007, media theorist Geert Lovink lectured and wrote about “the cool obscure” of new media art, and the panel discussion “Media Art Undone” took place at Transmediale in Berlin. On that occasion, artist Olia Lialina said: For a long time it did not make sense to show net art in real space: museums or galleries. For good reasons you had to experience works of net artists on your own connected computer. Yesterday for me as an artist it made sense only to talk to people in front of their computers, today I can easily imagine to apply to visitors in the gallery because in their majority they will just have gotten up from their computers. They have the necessary experience and understanding of the medium to get the ideas, jokes, enjoy the works and buy them.

Surfing clubs emerged in this transitional period, and mostly attracting young artists with an art education who, albeit critical of the art market and the process of commodification of digital artifacts, felt closer to the broader contemporary art discourse than to media art culture.

Explaining his approach to surfing and blogging in his essay “Spirit Surfing”, artist Kevin Bewersdorf naturally uses the work of Joseph Cornell as a reference point to show what happens when found online content is rearranged in a blog post. In October 2007, Bewersdorf had a show at And/Or Gallery in Dallas together with Guthrie Loneran, another influential pro-surfer. The exhibition featured videos, small prints, and a series of works that Bewersdorf produced by printing found online images onto ordinary objects like cushions, mouse pads and mugs using wall-greens.com, a web store with a print-on-demand service. Headed by artist and musician Paul Slocum, And/Or was a small artist-run space that, for a few years (from 2006 to 2009), gave the surfing club generation fertile terrain to present their work, test modes of presentation in the white cube without the constraints of a commercial gallery and the performance anxiety induced by a more central location, and build relationships with collectors. It anticipated many of the topics and tropes of post internet art, including the form of the “dual site”: a small, outlying, brick-and-mortar meeting point for an online community that emitted a signal amplified by the documentation on its web site, as Bewersdorf and Loneran did in their work, before they stopped producing visual art completely. For both, “retirement” started around 2009 – 2010, at the peak of their careers, and for very similar concerns about the objectification and commodification of their art. As Loneran said to Ed Halter in a recent interview:

Net art’s relationship to contemporary art as a whole and to the art market gets more confusing every day. It’s pretty fascinating to watch, though. I tried for years to figure out how to ‘print it out,’ to make something super-salable, but I could never quite figure it out, and I don’t think I ever will.

This short tour of the origins of post internet art should make it clear that it cannot be reduced to an art fair trend, to the small Bushwick scene described by Droitcour, or to the one liner “put a work on show, take a picture of it, and circulate the photo online”. On the contrary, the phenomenon seems to be the result of the confluence of a plurality of issues, approaches and processes, and the development of a new art community, less insular and more integrated than the one that developed around the first net art, and involved in an expanded conversation. At the risk of indulging in list making, we must at least mention artists like Hito Steyerl, Mark Leckey, Seth Price, and Metahaven, art magazines like Artforum, Mousse and Frieze, art critics and theorists like Boris Groys, David Joselit and Jennifer Allen, as part of or inspiration for this expanded conversation; of course alongside the good old crew, which was never dismissed or forgotten: When I started trying to be an artist in the early aughts, I most identified with the net art movement. I had discovered the work of jodi.org, Olia Lialina, Vuk Cosic, and



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Images gathered from “The New Aesthetic” Tumblr Archive

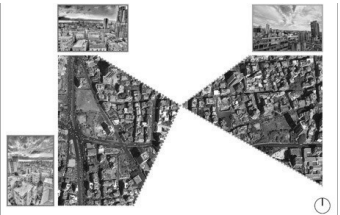
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Images
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from
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Aesthetic”
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Alexei Shulgin in college (outside of my studies) and I kinda had a “Ramones” moment, ... “I can do this!” It was terribly exciting and will seem obvious now, but the idea that one could just make something and people could see it without any intermediary was mind blowing. Also, for whatever reason – though one wonders if it was due to the lack of bandwidth (?) – the work of the above artists was surgically precise and conceptually clear headed. These traits undoubtedly added further fuel to my interest, and are things I still try to this day to emulate. At the time, if one was interested in browser art, by default they were linked to media art and its histories as the media scene was kinda the only game in town open for a dialog about the browser. This has all changed now as computers are mainstream therefore so is art which deals with them. This short tour of the origins of post internet art should make it clear that it cannot be reduced to an art fair trend, to the small Bushwick scene described by Droitcour, or to the one liner “put a work on show, take a picture of it, and circulate the photo online”. On the contrary, the phenomenon seems to be the result of the confluence of a plurality of issues, approaches and processes, and the development of a new art community, less insular and more integrated than the one that developed around the first net art, and involved in an expanded conversation. At the risk of indulging in list making, we must at least mention artists like Hito Steyerl, Mark Leckey, Seth Price, and Metahaven, art magazines like Artforum, Mousse and Frieze, art critics and theorists like Boris Groys, David Joselit and Jennifer Allen, as part of or inspiration for this expanded conversation; of course alongside the good old crew, which was never dismissed or forgotten:

When I started trying to be an artist in the early aughts, I most identified with the net art movement. I had discovered the work of jodi.org, Olia Lialina, Vuk Cosic, and Alexei Shulgin in college (outside of my studies) and I kinda had a “Ramones” moment, ... “I can do this!” It was terribly exciting and will seem obvious now, but the idea that one could just make something and people could see it without any intermediary was mind blowing. Also, for whatever reason – though one wonders if it was due to the lack of bandwidth (?) – the work of the above artists was surgically precise and conceptually clear headed. These traits undoubtedly added further fuel to my interest, and are things I still try to this day to emulate. At the time, if one was interested in browser art, by default they were linked to media art and its histories as the media scene was kinda the only game in town open for a dialog about the browser. This has all changed now as computers are mainstream therefore so is art which deals with them.

As we have seen, post internet art emerged at a time when net art was rehashing its relationship with the art world. Net art, which was always aware that its radical autonomy could only be temporary, and that always played the tongue in cheek role of “the last avant-garde”, was nonetheless initially ‘mind blown’ (in the words of Arcangel) by the opportunity to play different roles (including those of the institution and gallery) and reach an audience “without any intermediary”. Yet, it soon understood that in order to make the game more interesting, the art world was not to be rejected, but used and abused. This is what Natalie Bookchin and Alexei Shulgin, in their seminal piece Introduction to net.art (1994-1999), call “the cultural loop”:

- 2. Interface with Institutions: The Cultural Loop
 - a. Work outside the institution
 - b. Claim that the institution is evil
 - c. Challenge the institution
 - d. Subvert the institution
 - e. Make yourself into an institution
 - f. Attract the attention of the institution
 - g. Rethink the institution
 - h. Work inside the institution

What should be understood about this process is that it’s a loop, not a straight line. Thus, “work inside the institution” is not the end of a process that goes from autonomy to integration, but part of a continuous loop in which net celebrity serves to get into the gallery, and art world recognition serves to generate more interest online. The gallery and the museum are not the end point. And integration is not the end point, but a means to get more autonomy. This does not mean, of course, that net art has to have a co-dependent relationship with the system. There is a long list of artists who developed substantial bodies of work and were extremely influential among their peers, but who refused to join the cultural loop: Bewersdorf and Loneragan are good examples. But anyone interested in a career in art can’t avoid joining this level of the game. A good example of this comes from the artists involved in the Free Art & Technology Lab (F.A.T. Lab), a group co-founded by Evan Roth and James Powderly in 2007 during an Eyebeam residency, and now numbering 20+ artists, designers and hackers from three continents. [32] F.A.T. Lab not only includes “professional artists”, but also professionals from various different disciplines, interested in having an arena for experimentation where they can work on projects that elude easy categorization, and can interest different audiences. Artists associated with F.A.T. Lab include Evan Roth, Aram Bartholl, Addie Wagenknecht, Golan Levin, among others. They all have a respected institutional presence, and they all have gallery representation. Some of their works could easily

fit into one of the definitions of post internet art; but they keep working with technology, making online projects, participating in group projects, cultivating relationships with communities outside of the realm of art: hackers, graffiti artists, the open software and open hardware movements. The art world is just one mode of transmission, and maybe not even the most important one. Their artworks contribute to developing an identity that is much richer than those of the art world artists we are used to (perhaps closer to what an artist should be, and the best artists have been).

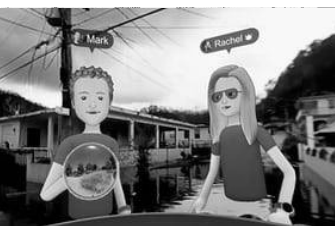
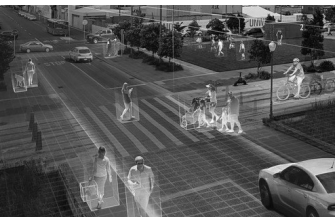
In the context of this cultural loop, the posited opposition between the radical independence of net art and post internet art's "obsession with art-world power systems" (Droitcour) shows its flaws. Let's take post internet's most criticized feature: its use of the gallery as a stage for taking documentary pictures to be distributed online. Droitcour writes:

The Post-Internet art object looks good in a browser just as laundry detergent looks good in a commercial. Detergent isn't as stunning at a laundromat, and neither does Post-Internet art shine in the gallery. It's boring to be around. It's not really sculpture. It doesn't activate space. It's often frontal, designed to preen for the camera's lens. It's an assemblage of some sort, and there's little excitement in the way objects are placed together, and nothing is well made except for the mass-market products in it. It's the art of a cargo cult, made in awe at the way brands thrive in networks.

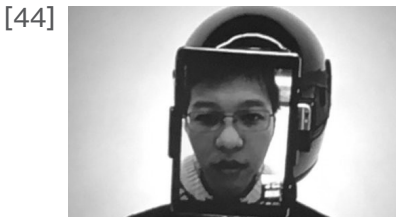
Let's suppose this is true. After all, artists associated with post internet often claim to view mediated experience on the same level as primary experience (Laric), and to do visual art for the sake of Facebook likes (Ito); and Artie Vierkant based all his Image Objects project on the circular relation(another loop) between physical object and digital documentation. They all seem to agree with David Joselit when he states that value and aura, today, are not generated by uniqueness and geographical specificity, but by replicability and ubiquity. In our framework, this only means that the physical artifact is not the artwork, but part of a (mostly immaterial) process in which the physical object is just one step; and that the white cube is not the main space in which the artwork manifests itself, but one of the contexts in which the art is created. You can't blame something for not being what it doesn't want to be, and for not doing what it doesn't want to do. Of course, the fact of creating a traditional, physical artifact and displaying it in the white cube seems to sanction the traditional systems of attribution of aura: turn your art into a commodity and place it in an art space. But how much criticism is there in this process of (subversive) affirmation?

In 1964, Yoko Ono published an artist book called Grapefruit, containing a series of "event scores", instructions that replace the physical work of art. One of them is titled "PAINTING TO EXIST ONLY WHEN IT'S COPIED OR PHOTOGRAPHED", and reads: "Let people copy or photograph your paintings. Destroy the originals." At the core of this piece there is the definitive traditional art object, a painting. The "score" does not need to, but legitimately can, be performed: if so, the final result (the copy, the photograph) will have the same status as the idea. It's conceptual art at its best, claiming the supremacy of ideas over artworks-as-commodities, and combating the notion of originality. Here, the painting, the copy and the photograph are part of a process; nobody cares if they are good or bad; what's important is the process they are part of, and its meaning. In 2013, artist Joshua Citarella curated what looks like a good post internet reenactment of Ono's event score. Compression Artifacts was a set emulating all the features of the white cube (white walls, gray floor, neon lighting), built in an undisclosed location in a forest, and performed in front of a live feed, broadcast during daylight hours. Once it was completed, works by Wyatt Niehaus, Kate Steciw, Brad Troemel, Artie Vierkant and Joshua Citarella were set up in the space. Following the exhibition the structure was demolished. All artworks and building materials were incinerated and regathered as ash. And all the process was, of course, documented in pictures. Compression Artifacts is probably the most direct manifestation of the post internet as a subversive affirmation. The sacrality of the physical artifact and the white cube are first sanctioned, only to be demystified: the artworks are burnt, the white cube is built in a forest (and then burnt). The process was only visible on the internet, first as streaming video, then as documentation. The documentation itself is not the artwork, not the end point: it's just a way to keep the ball rolling.

Post internet art – at least, that post internet art that puts emphasis on the importance of mediation and distribution – exists in the white cube as the artworks displayed in Compression Artifacts, as the temporary materialization of an idea (that may eventually be destroyed, or sold to a collector). What's important is not the piece in there, but the idea out there. This idea does not manifest itself as a single object, but is most effectively exemplified by the digital image. It is free, it travels, it gathers metadata along the way, it can be appropriated, used, abused, perused, and further developed. It can show up in different contexts. It's ephemeral, but it can survive.



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The Web Spilled Over

“Is the Internet Dead?”

Hito Steyerl. “Too Much World: Is the Internet Dead?” (I Am a Minecraft RedstoneComputer, Postproduction), The Internet Does Not Exist, 2015



Is the internet dead? This is not a metaphorical question. It does not suggest that the internet is dysfunctional, useless or out of fashion. It asks what happened to the internet after it stopped being a possibility. The question is very literally whether it is dead, how it died and whether anyone killed it.

But how could anyone think it could be over? The internet is now more potent than ever. It has not only sparked but fully captured the imagination, attention and productivity of more people than at any other point before. Never before have more people been dependent on, embedded into, surveilled by, and exploited by the web. It seems overwhelming, bedazzling and without immediate alternative. The internet is probably not dead. It has rather gone all-out. Or more precisely: it is all over! This implies a spatial dimension, but not as one might think. The internet is not everywhere. Even nowadays when networks seem to multiply exponentially, many people have no access to the internet or don't use it at all. And yet, it is expanding in another direction. It has started moving offline. But how does this work?

Remember the Romanian uprising in 1989, when protesters invaded TV studios to make history? At that moment, images changed their function. Broadcasts from occupied TV studios became active catalysts of events—not records or documents. Since then it has become clear that images are not objective or subjective renditions of a preexisting condition, or merely treacherous appearances. They are rather nodes of energy and matter that migrate across different supports, shaping and affecting people, landscapes, politics, and social systems. They acquired an uncanny ability to proliferate, transform, and activate. Around 1989, television images started walking through screens, right into reality. This development accelerated when web infrastructure started supplementing TV networks as circuits for image circulation. Suddenly, the points of transfer multiplied. Screens were now ubiquitous, not to speak of images themselves, which could be copied and dispersed at the flick of a finger.

Data, sounds, and images are now routinely transitioning beyond screens into a different state of matter. They surpass the boundaries of data channels and manifest materially. They incarnate as riots or products, as lens flares, high-rises, or pixelated tanks. Images become unplugged and unhinged and start crowding off-screen space. They invade cities, transforming spaces into sites, and reality into realty. They materialize as junkspace, military invasion, and botched plastic surgery. They spread through and beyond networks, they contract and expand, they stall and stumble, they vie, they vile, they wow and woo.

Just look around you: artificial islands mimic genetically manipulated plants. Dental offices parade as car commercial film sets. Cheekbones are airbrushed just as whole cities pretend to be YouTube CAD tutorials. Artworks are e-mailed to pop up in bank lobbies designed on fighter jet software. Huge cloud storage drives rain down as skylines in desert locations. But by becoming real, most images are substantially altered. They get translated, twisted, bruised, and reconfigured. They change their outlook, entourage, and spin. A nail paint clip turns into an Instagram riot. An upload comes down as shitstorm. An animated GIF materializes as a pop-up airport transit gate. In some places, it seems as if entire NSA system architectures were built—but only after Google-translating them, creating car lofts where one-way mirror windows face inwards. By walking off-screen, images are twisted, dilapidated, incorporated, and reshuffled. They miss their targets, misunderstand their purpose, get shapes and colors wrong. They walk through, fall off, and fade back into screens.

Grace Jones's 2008 black-and-white video clip “Corporate Cannibal,” described by Steven Shaviro as a pivotal example of post-cinematic affect, is a case in point. By

now, the nonchalant fluidity and modulation of Jones’s posthuman figure has been implemented as a blueprint for austerity infrastructure. I could swear that Berlin bus schedules are consistently run on this model—endlessly stretching and straining space, time, and human patience. Cinema’s debris rematerializes as investment ruins or secret “Information Dominance Centers.” But if cinema has exploded into the world to become partly real, one also has to accept that it actually did explode. And it probably didn’t make it through this explosion either.

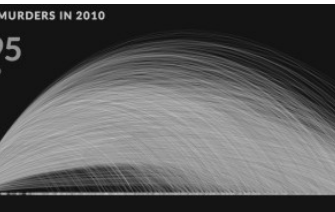
I AM A MINECRAFT REDSTONE
COMPUTER

So what does it mean if the internet has moved offline? It crossed the screen, multiplied displays, transcended networks and cables to be at once inert and inevitable. One could imagine shutting down all online access or user activity. We might be unplugged, but this doesn’t mean we’re off the hook. The internet persists offline as a mode of life, surveillance, production, and organization – a form of intense voyeurism coupled with maximum nontransparency. Imagine an internet of things all senselessly “liking” each other, reinforcing the rule of a few quasi-monopolies. A world of privatized knowledge patrolled and defended by rating agencies. Of maximum control coupled with intense conformism, where intelligent cars do grocery shopping until a Hellfire missile comes crashing down. Police come knocking on your door for a download – to arrest you after “identifying” you on YouTube or CCTV. They threaten to jail you for spreading publicly funded knowledge? Or maybe beg you to knock down Twitter to stop an insurgency? Shake their hands and invite them in. They are today’s internet in 4D.

The all-out internet condition is not an interface but an environment. Older media as well as imaged people, imaged structures, and image objects are embedded into networked matter. Networked space is itself a medium, or whatever one might call a medium’s promiscuous, posthumous state today. It is a form of life (and death) that contains, sublates, and archives all previous forms of media. In this fluid media space, images and sounds morph across different bodies and carriers, acquiring more and more glitches and bruises along the way. Moreover, it is not only form that migrates across screens, but also function. Computation and connectivity permeate matter and render it as raw material for algorithmic prediction, or potentially also as building blocks for alternate networks. As Minecraft Redstone computers are able to use virtual minerals for calculating operations, so is living and dead material increasingly integrated with cloud performance, slowly turning the world into a multilayered motherboard.



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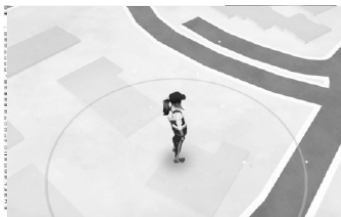


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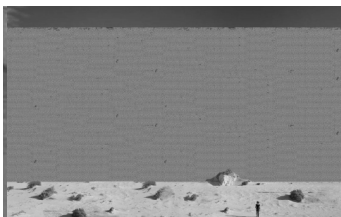
[46] [47]
Images gathered from “The New Aesthetic” Tumblr Archive



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[51] [52] [53]
Images gathered from “The New Aesthetic” Tumblr Archive

But this space is also a sphere of liquidity, of looming rainstorms and unstable climates. It is the realm of complexity gone haywire, spinning strange feedback loops. A condition partly created by humans but also only partly controlled by them, indifferent to anything but movement, energy, rhythm, and complication. It is the space of the ronin of old, the masterless samurai freelancers fittingly called wave men and women: floaters in a fleeting world of images, interns in dark net soap lands. We thought it was a plumbing system, so how did this tsunami creep up in my sink? How is this algorithm drying up this rice paddy? And how many workers are desperately clambering on the menacing cloud that hovers in the distance right now, trying to squeeze out a living, groping through a fog which may at any second transform both into an immersive art installation and a demonstration doused in cutting-edge tear gas?

POSTPRODUCTION

But if images start pouring across screens and invading subject and object matter, the major and quite overlooked consequence is that reality now widely consists of images; or rather, of things, constellations, and processes formerly evident as images. This means one cannot understand reality without understanding cinema, photography, 3D modeling, animation, or other forms of moving or still image. The world is imbued with the shrapnel of former images, as well as images edited, photoshopped, cobbled together from spam and scrap. Reality itself is postproduced and scripted, affect rendered as after-effect. Far from being opposites across an unbridgeable chasm, image and world are in many cases just versions of each other. They are not equivalents however, but deficient, excessive, and uneven in relation to each other. And the gap between them gives way to speculation and intense anxiety. Under these conditions, production morphs into postproduction, meaning the world can be understood but also altered by its tools.

The tools of postproduction: editing, color correction, filtering, cutting, and so on are not aimed at achieving representation. They have become means of creation, not only of images but also of the world in their wake. One possible reason: with digital proliferation of all sorts of imagery, suddenly too much world became available. The map, to use the well-known fable by Borges, has not only become equal to the world, but exceeds it by far. A vast quantity of images covers the surface of the world – very in the case of aerial imaging – in a confusing stack of layers. The map explodes on a material territory, which is increasingly fragmented and also gets entangled with it: in one instance, Google Maps cartography led to near military conflict.

While Borges wagered that the map might wither away, Baudrillard speculated that on the contrary, reality was disintegrating. In fact, both proliferate and confuse one another: on handheld devices, at checkpoints, and in between edits. Map and territory reach into one another to realize strokes on trackpads as theme parks or apartheid architecture. Image layers get stuck as geological strata while SWAT teams patrol Amazon shopping carts. The point is that no one can deal with this. This extensive and exhausting mess needs to be edited down in real time: filtered, scanned, sorted, and selected – into so many Wikipedia versions, into layered, libidinal, logistical, lopsided geographies. This assigns a new role to image production, and in consequence also to people who deal with it. Image workers now deal directly in a world made of images, and can do so much faster than previously possible. But production has also become mixed up with circulation to the point of being indistinguishable. The factory/studio/tumblr blur with online shopping, oligarch collections, realty branding, and surveillance architecture.

Today's workplace could turn out to be a rogue algorithm commandeering your hard drive, eyeballs, and dreams. And tomorrow you might have to disco all the way to insanity. As the web spills over into a different dimension, image production moves way beyond the confines of specialized fields. It becomes mass postproduction in an age of crowd creativity. Today, almost everyone is an artist. We are pitching, phishing, spamming, chain-liking or mansplaining. We are twitching, tweeting, and toasting as some form of solo relational art, high on dual processing and a smartphone flat rate. Image circulation today works by pimping pixels in orbit via strategic sharing of wacky, neo-tribal, and mostly US-American content. Improbable objects, celebrity cat GIFs, and a jumble of unseen anonymous images proliferate and waft through human bodies via Wi-Fi. One could perhaps think of the results as a new and vital form of folk art, that is if one is prepared to completely overhaul one's definition of folk as well as art. A new form of storytelling using emojis and tweeted rape threats is both creating and tearing apart communities loosely linked by shared attention deficit.

[54]



[55]



[56]



[54] [55]
[56]
Images gathered from “The New Aesthetic” Tumblr Archive

Thomas Ruff (2007)

(MoMA) Collection. Retrieved 12/11/2025



Thomas Ruff produced his Jpegs series without using a camera. Instead, he printed compressed electronic image files (jpegs), many of them downloaded from the Internet, at monumental scale. The series, which encompasses tranquil landscapes, scenes of war, and natural and man-made catastrophes, began with images of the September 11th terrorist attacks. To make this pixelated image of an erupting volcano, Ruff printed a low-resolution jpeg at a very large scale. At such size, the pixels—the building blocks of all digital images—are magnified to the point of breaking up the very image they compose. By foregrounding the repurposed photograph's underlying structure and naming (msh for Mount St. Helens), Ruff calls attention to the ways digital images are constructed, circulated, and viewed today. [B]

jpegs

Everything about “The New Aesthetic”

“Tediously Academic Essay
on the New Aesthetic”

Katja Kwastek. Thinking Postdigital Aesthetics: Art, Computation and Design. (Howto Be Theorized: A Tediously Academic Essay on the New Aesthetic, 2015)

SOMTHING WITH A NAME

On 6 May 2011, James Bridle— who self-identifies as ‘a writer, publisher, technologist, and artist, based in London’²— launched a Tumblr blog entitled The New Aesthetic, featuring an initial 25 entries.³ Over the course of the following year, new entries were added to the Tumblr on a regular basis, between 100 and 150 each month. Parallel to this, Bridle presented his concept in several lectures.

But the New Aesthetic only really took off in spring 2012, when it was the topic of a panel at the South by Southwest conference, which was reviewed for the popular online magazine WIRED.com by science fiction author Bruce Sterling (2012). From this moment on, Twitter messages and blog entries referring to the New Aesthetic increased tremendously (see, as an example, Kaganskiy 2012). In parallel, a number of further conferences and festival events on the topic were organized, including a book sprint. Even while the initial Tumblr blog was (temporarily) closed, the concept continued to spread across the digital networks and has been related to quite diverse visual and objectual, material and immaterial artefacts— in blog posts and Twitter messages all over.

[57][58]
Images
gathered
from
“The New
Aesthetic”
Tumblr
Archive



FIG. 8A



The New Aesthetic thus underwent a kind of viral disper-
sion– it resembles an internet meme (see Watz 2012),
although with a proper name. As Bridle (2012) puts it,
‘giving something a name gives you power over it’. How-
ever, he also flirts with emphasizing that the denomination
‘sounds more portentous than I mean’ (2011a) or with
calling it ‘a rubbish name [...which] seems to have taken
hold’ (2011b). While Bridle is unanimously acknowledged
as something like the guiding spirit of the New Aesthet-
ic, the question of what it actually is that he is credited
with having baptized remains unclear to the present day.
As of today (July 2014), the New Aesthetic Tumblr blog
contains more than 2,800 entries. It collocates images of
design objects as well as street art; it features pixelated
patterns as well as references to game aesthetics; it con-
tains satellite images, data visualizations glitchy digital
images, documentations of camouflage markings, and
many other things.

Attempts to characterize the collection of items present-
ed under the heading of the New Aesthetic range from
‘crowdsourced heaps of eye-candy’ (Sterling 2012) via a
‘major trend in design’ (Sterling 2012) to a ‘conceptual art
movement’ (Minard 2012). Scholars relate it to concepts
as diverse as remediation (Grusin 2012), computa-
tionality (Berry 2012), object-oriented ontology (Borenstein
2012) and post-internet art (Wallace 2014). Bruce Sterling
even addresses it as an avant-garde movement (Sterling
2012). However, Bridle (2013) vehemently objects to the
denomination of the New Aesthetic as a movement: ‘the
New Aesthetic may be considered a work, a conversation,
a performance, an experiment, and a number of other things
(although, please, not a movement)’.

I agree that characterizing the New Aesthetic as a move-
ment is inappropriate, because it unites neither a group
of actors or artists with a common intention, nor a group
of works or items featuring a common style, nor a series
of connected innovations. Bridle’s collection contains
items which are very heterogeneous in terms of medial
status, intentionality of production, level of reflection and
target audience. While there might be a common de-
nominator, there is no common thingness, or ontological
status. This gives plausibility to Bridle’s statement on the
Tumblr blog that he actually wants the New Aesthetic to
be understood as ‘an ongoing research project’. This has
led to critique by Sterling (2012), who disavows Bridle’s
project as a ‘valiant cut-and-paste campaign’, which might
look like ‘traditional criticism, but is actually blogging and
tumblr-ing’. However, in a later blog post, Bridle defends
his position by advocating the specific format of the New
Aesthetic– which differs from traditional academic formats
of print culture as a distinguishing feature. He sees it as
an attempt to “write” critically about the network in the
vernacular of the network itself’ (Bridle 2013). This said,

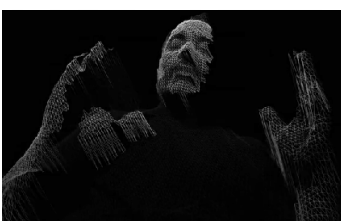
it becomes clear that any serious discussion of the New
Aesthetic has to take into account both its object(s) of
interest and the way it addresses and presents them. Let
us, nonetheless, start with the former.

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WAVING AT THE MACHINES WITH A PIXELATED FINGER

To elucidate his object(s) of interest, Bridle offered var-
ious catchy metaphors. The first was introduced in the
first lecture he gave on the topic of the New Aesthetic.
He entitled it ‘Waving at the Machines’ (Bridle 2011b). By
alluding to the Kinect ritual of having to raise one’s arms
in order to be identified by the system, Bridle presented
a compelling image for the new computer vision-based
human–computer interaction (HCI) technologies which
can ‘read’ our gestures: humans can control technoloical
systems by means of bodily movement, or– to put it the
other way round– the everyday becomes increasingly
machine readable. However, most computer-vision en-
tries mentioned in Bridle’s talk and collected in the Tumblr
actually concern attempts to prevent machine readability.
This goes for different forms of camouflage patterns on
ships and airplanes as well as for Adam Harvey’s CV
Dazzle project. Such items present means of cheating
the machines, as opposed to waving at them.

[59]



[60]



[59] [60]
Images
gathered
from
“The New
Aesthetic”
Tumblr
Archive

The second, much broader and fuzzier metaphor claims ‘eruptions of the digital into the physical world’. While computer vision is quite specific as a technology, the idea of an increasing pervasiveness of the digital throughout everyday life allows for a broad spectrum of interpretations. It can be related both to actual computational processes and to visual artefacts which make the ongoing computation visible in the context of our physical environment and material culture. Bridle (2012) described such ‘glitchy and pixelated imagery’ as a ‘useful visual shorthand’ for ‘collaboration with technology’. He sees such imagery as a ‘look’, which is a ‘metaphor for understanding and communicating the experience of a world in which the New Aesthetic is increasingly pervasive’. However, the New Aesthetic was critiqued for its obvious focus on the visual appearance and effects of technology. Accordingly, the supporters of the term hastened to emphasize that the New Aesthetic artefacts had to be understood as metaphors for what lies underneath, and the New Aesthetic had to be seen as an enquiry into the objects which generate the respective images (Borenstein 2012). In a 2013 essay, Bridle even claimed that the New Aesthetic is ‘concerned with everything that is not visible in these images and quotes, but that is inseparable from them, and without which they would not exist’ (Bridle 2013). He seems to regret having used the term ‘Aesthetic’ at all, and argues that critics who have looked only at the surface are mistaken. Resorting to another metaphor, he complains that critics have looked ‘at the pixelated finger, not the moon’. I agree that dismissing the New Aesthetic as superficial is inappropriate, but not because I would approve the claim that it ultimately concerns the invisible. On the contrary, I hold that it is the visible— or, as I shall explain shortly, the perceivable— which gives heart to the very notion. To use Bridle’s metaphor, it is the pixelated finger, and not the moon, which interests me here.

The concept of a New Aesthetic can be highly valuable if we use it to investigate how digital technologies shape the perceivable world— and our perception of the latter. If we want to discuss what lies beneath, to analyse the processes these artefacts and images (re)present, we actually do not need a new term, because we have whole academic disciplines devoted to such research (media studies, science and technology studies, etc.), and we can draw on a set of very specific concepts to differentiate between the various technologies and developments at stake (ubiquitous computing, the Internet of Things, computer vision, etc.). I hold that the notion of a New Aesthetic only makes sense if we take its focus on aesthetics seriously. Aesthetics as a philosophical notion has been used in various ways, but it has always been related to the perceivable. The focus may lie on the perceivable item and on its qualities or attraction. But aesthetics may equally well denote a theory of visuality, or be used to

scrutinize the act of perception. Addressing aesthetics does not equal being superficial or lacking depth. Rather, it entails a shift of focus. Instead of mainly discussing what is represented, an aesthetic approach asks how something is represented and perceived— though, of course, both aspects are deeply intertwined.

(RE)MEDIATION

When I first watched the online recording of James Bridle’s talk at the 2011 Web Directions South conference, I immediately related many of his examples to the concept of ‘remediation’ as introduced by Jay David Bolter and Richard Grusin. They define remediation as ‘the representation of one medium in another’ (Bolter and Grusin 2000, p. 45). So if we think of Helmut Smits’ Dead Pixel, emulating a bitmap error in the physical landscape, of voxelated sculptures, of pixelated fabric or polygon-shaped shoes, we can see these as digital media being represented on— or remediated to— real-world surfaces. In reverse, ‘render ghosts’ are portraits of ordinary people inhabiting digital simulations. However, Bolter and Grusin mainly coined the term to discuss progressive forms of media use, within which new media adopt formal characteristics of older ones to increase their usability. In Bolter and Grusin’s book, we can see a strong tendency towards an understanding of ‘remediation as reform’, on the basis of the assumption of a constant improvement, or at least evolution, of media. Although they mention that ‘hypermedia applications [as...] acts of remediation [...] import earlier media into a digital space in order to critique and refashion them’, the idea of remediation as reflection or iconic representation is not central to Bolter and Grusin. This becomes quite clear in Grusin’s (2012) Twitter comment on the New Aesthetic, arguing that ‘The “new aesthetic” is just the latest name for remediation, all dressed up with nowhere to go.’ While the concept of remediation is a functional one, ‘going somewhere’ does not seem to be the main goal of many New Aesthetic artefacts— at least concerning the examples mentioned above. Rather, they highlight by decontextualizing, mostly in terms of a backwards mediation, from the digital back to the analogue, from the virtual back to the physical, abstracting a visual language from its technological conditions.

However, as I have shown elsewhere, relating these artefacts to the concept of remediation helps to historicize them, to show that inter-media references, as well as the interdependence of visual styles and technological developments, are nothing new (Kwastek 2015). At the same time, the deficiency of the concept of remediation to fully apprehend the New Aesthetic urges us to further unfold the different strategies of mediation at stake, especially if we want to do justice to the full range of items

[61] [62]
[63]
Images gathered from “The New Aesthetic” Tumblr Archive



presented under its heading. the New Aesthetic urges us to further unfold the different strategies of mediation at stake, especially if we want to do justice to the full range of items presented under its heading. If we scan the Tumblr collection a bit more carefully, next to items which transfer or remediate digital assets to the material world (3D prints, voxel sculptures), or images of objects which appropriate styles originating from digital technologies (pixelated patterns), we can distinguish entries that capture, highlight or just represent digital artefacts (glitches, error messages, Street View snippets, but also ‘serious’ technological inventions), and items which manipulate, remix, mash up, hack or misuse ‘digital found footage’ (game hacks, data-moshes).

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The whole Tumblr collection is highly heterogeneous, and, if we were to try to cluster it, the resulting visualization would show an increasing fuzziness towards the edges of the field, including mere associative similarities and textual comments on various sorts of new technologies. However, in a way it seems that the fuzziness of this associative collection is at the core of the project, because it concerns both the collection itself and the way it is presented, developed, discussed and distributed. Further,

the project can serve as evidence for a new messiness of media which characterizes our contemporary culture in general. Its setup constitutes a perfect example of what Clay Shirky– in reference to textual publication– has termed the ‘publish, then filter’ paradigm (2008, chapter 4): as online publication is cheap and simple, why not publish as many voices as exist, and leave it to the reader to evaluate them? Concerning visual culture, the New Aesthetic seems to call for what Lev Manovich advocates with his concept of exploratory data analysis (Manovich 2013, 257–258): if one wants to arrive at really new and surprising insights, one should start by visualizing a data collection without too much of a restricting question or hypothesis in mind. However, concerning a concept as fuzzy as the New Aesthetic, leaving the filtering to the audience entails that everybody can come up with their own ‘New Aesthetic’ on the basis of a choice of examples which best fits their individual interest. And any such attempt at pinning the concept down can be countered by quoting some contradictory examples, which will certainly also feature within the huge number of items published. Accordingly, while a discussion in terms of (re)mediation is helpful in discussing some of the items grouped under the heading of the New Aesthetic, it does not do justice to all of them by far. David Berry objects to the contextualization of the New Aesthetic as a form of remediation for another reason. He argues that the concept of remediation perpetuates the notion of medium specificity and a hegemony of the visual. It does not take into account a key characteristic of digital media, which is computability. Berry (2012) argues that we have to ‘think in terms of a medium as an agency or means of doing something’, which also entails ‘thinking beyond the screenic’. Thinking along this line, the concept of remediation, as introduced by Bolter and Grusin, attempts to create interfaces for computational media, using metaphors of older media to make the computational intelligible for and operable by humans. The New Aesthetic, in contrast, is arguably all about computational media, but quite a few of its examples derive their attraction from actually being fully analogue, material or physical. Again, this is true only for a fraction of the items presented. Interactivevisualizations are computational, and so are computer-vision technologies.

But, when it comes to glitchy imagery, we already have to differentiate. Glitches are the results of computation, but, as long as they are not (real time) generative, they are not computational, but mere documents of prior computational errors. Pixelated or polygon-shaped patterns refer to computational processes, but they are not even processual, let alone computational. We could even argue that, once new media have been remediated or backwards mediated to material culture, they lose their computability and are reduced to static images and artefacts,

[64] [65]
[66]
Images gathered from “The New Aesthetic” Tumblr Archive



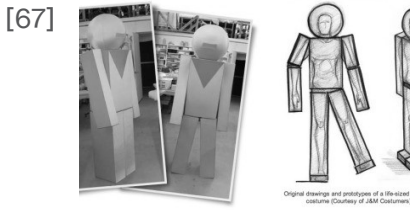
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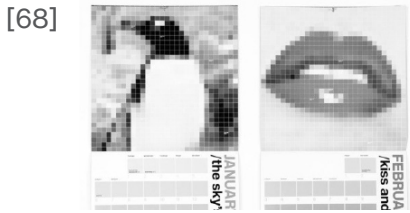
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Images gathered from “The New Aesthetic” Tumblr Archive

while referencing computability by means of abstracted visual properties. They become mere icons of the digital. But is this really true? Might the ultimate merit of the New Aesthetic project not lie in the fact that it questions this very assumption, that it brings into question not only the clear distinction between the digital and the material, but also the distinctiveness of the computational as a clearly defined category? The New Aesthetic started with a strong focus on machine readability of the everyday. But, if the everyday has become machine readable, do we not have to question our concept of the computational? Some visual artefacts have been produced so as to be machine readable from the outset, like the barcode. Though it is not electronic, it is machine readable, and contains computable code. But what many of the New Aesthetic artefacts point to, even if often rather iconically or metaphorically, is the fact that we do not even have to wave at the machines to be captured by their computer-vision systems in order to have our images or movements analysed and processed. Thanks to computer-vision technology, the whole world is becoming potentially computational, or, as German artist Hito Steyerl (2014, p. 34) puts it, matter becomes ‘raw material for algorithmic prediction’, and ‘living and dead material [is] increasingly integrated with cloud performance’.

NEWNESS VERSUS POTENTIALITY

Both concepts, the New Aesthetic and the post-digital, operate with a time based characterization. They both position themselves in relation to some kind of assumed linear development— while the one claims to ‘be new’, the other claims to ‘come after’. But the prefix ‘post’ has come to be used not only to signal the end of one period and its displacement by another, but also to denote rather ‘subtle cultural shifts and ongoing mutations’ (Cramer 2014, reprinted in this volume). We apply the prefix ‘post’ to question some established concept which we might have all too easily taken for granted. It has come to emphasize a need for differentiation and critical review, a scepticism about the universal validity of a certain concept.

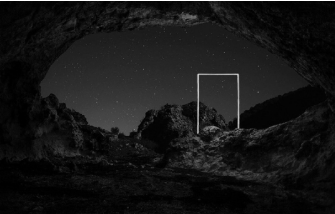
While the term ‘post-digital’ was initially coined in the context of musicology (Cascone 2002), it has by now become a prompt for a critical review of the role of digital technologies in our contemporary society. Florian Cramer sees it as signalling ‘a contemporary disenchantment with digital information systems and media gadgets’ (Cramer 2014, reprinted in this volume). But, more specifically, the notion of the post-digital is used to acknowledge that, today, digital technology is deeply embedded in ‘everyday life’. It serves to emphasize that ‘the digital’ is not as definite as we might assume: that it is no ‘virtual reality’ distinct from our everyday world, but a constitutive part

of it. This actually sounds very familiar and quite close to the idea of ‘eruptions of the digital into the physical world’. Accordingly, the New Aesthetic is ultimately no more and no less than a post-digital aesthetics an investigation of the effects of the increasing merging of the digital and the material, with the focus on its perceivable effects and our perceptive formation. However, comparing the two terms, there is a slight difference in connotation, which results mainly from the differing choice of time-based qualifiers, but might be symptomatic of the respective research perspectives taken: while postness insinuates some kind of reflective distance and disenchantment, newness implies a considerable amount of fascination, or at least, wonder. The New Aesthetic Tumblr certainly includes highly reflective and critical items, but the project as a whole is rather neutral in terms of a position to take, even if we take into account the lectures and online essays related to it. It mainly presents findings detected through surfing the web, which, in turn, allows for its great diversity and heterogeneity. It presents its objects of interest largely without classifying, judging or theorizing them. If we follow Bridle’s argumentation, this actually constitutes an important characteristic of the whole concept. As mentioned, he wants it to be writing ‘critically about the network in the vernacular of the network itself’. Obviously the fascination with the new, the iconic, the curiosity, the meme, the quick statement and its broad dissemination is characteristic of this ‘vernacular’. That is why writing critically in this very language obviously constitutes a big challenge, if it is not in terms of traditional scholarly blog entries that merely complement the Tumblr and Twitter statements. But it is not impossible.

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Images gathered from “The New Aesthetic” Tumblr Archive



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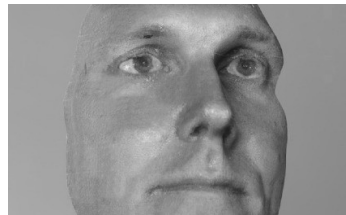


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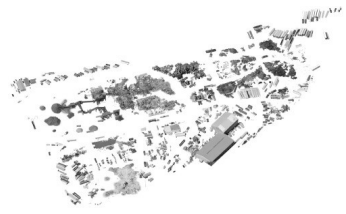
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Images gathered from “The New Aesthetic” Tumblr Archive

James Bridle (2011)

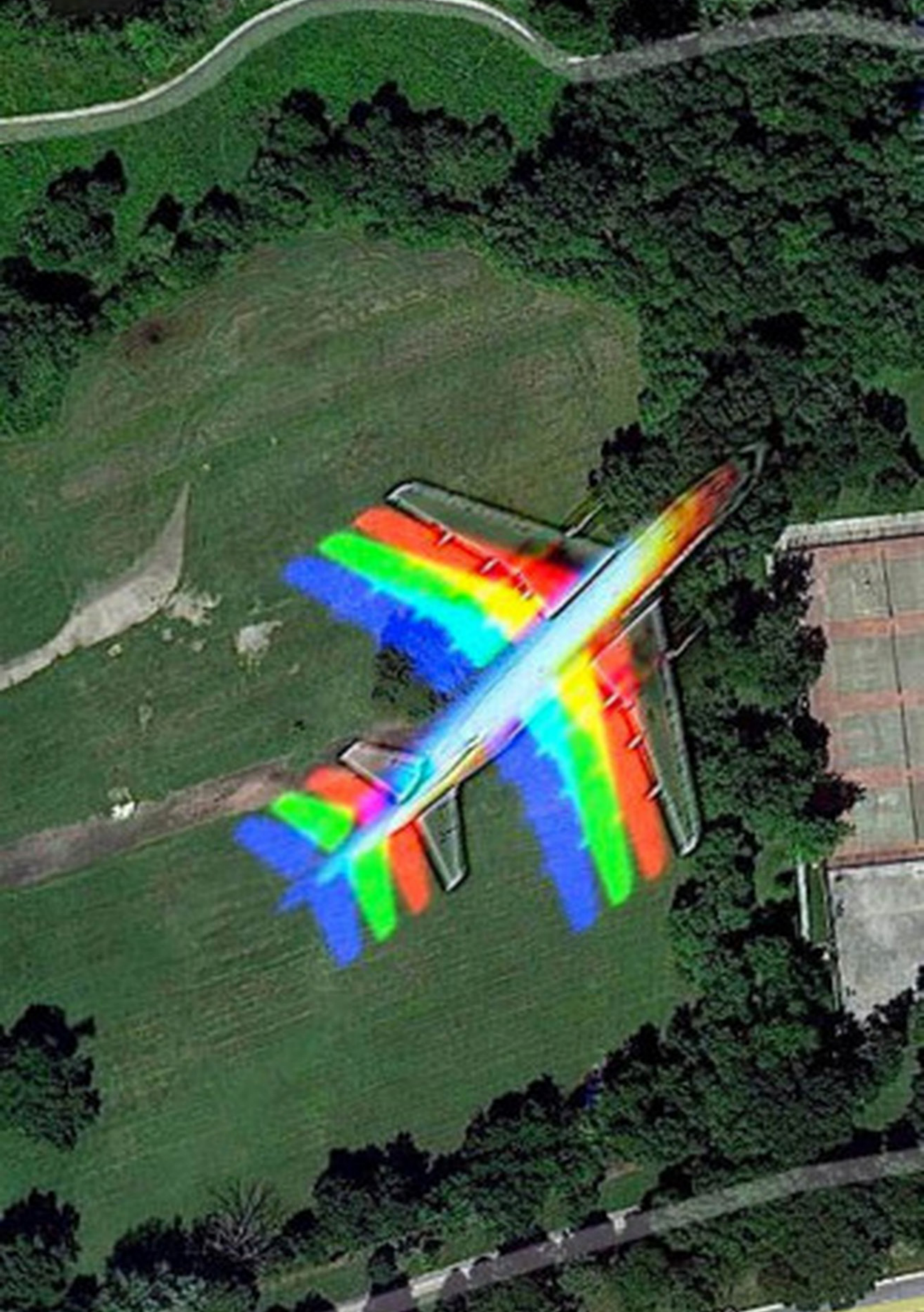
[C] Bridle, J. (2011). The New Aesthetic.

Retrieved 12/11/2025

“The New Aesthetic”

As fascinating as its name is bland, “the New Aesthetic” refers to a website launched in 2011 by the London-based artist, designer, and writer James Bridle; to the many images archived on that site; to Bridle’s argument about the sensibility that those images share; to the international panoply of pictures, websites, texts, and computer programs created in response; and to a heated debate about what they mean. The New Aesthetic instructs us to look to the future, but it might be most interesting for what it says about the recent past.

Put too simply, the New Aesthetic is an art movement, or an attempt to launch one, that welcomes human collaborations with real or imagined digital machines. The earliest pictures on Bridle’s site were either real objects that looked like crude programmed creations, or digital images whose oddball features put their difference from “real life” up front. Coarse green squares in an arid, tan landscape looked like early-’80s Atari graphics but were really “agricultural land seen from space.” A pendulous, blue-on-blue column of LEGO-like blocks, representing water, “poured” out from a New York City pipe. Later works included imaginary buildings shaped like QR codes, an essay about Auto-Tune, and much reuse of Google Street View.day. [C]

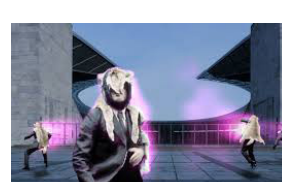
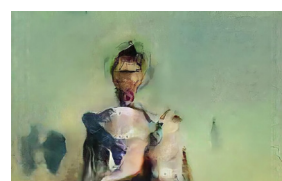
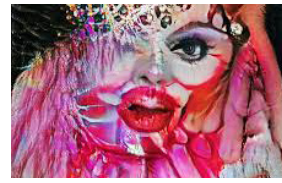


Ian Cheng, Emissary In The Squat Of Gods,
live simulation and story, infinite duration, 2015.



Animal Spirit Hito Steyerl (2022)	Generative Culture
Neural Zoo Sofia Crespo (2019-2021)	Generative Art
Machine Hallucinations Refik Anadol (2019)	Made By and For Humans?
Portraits of Imaginary People Mike Tyka (2019)	Amidst the Uncanny
	Eating the Future

Where We Are Going



**"Aesthetic products,
a collective and shared
culture-building phenomena
would be compromised by
excessive individualization
of the content each person
would consume by means
of tailored AI-generation."**

Generative Culture

“AI and Production
of Culture”

Manovich, L. AI Aesthetics. Strelka Press. (“AI
and Production of Culture”, pp. 7-18), 2018



AI plays a crucial role in the global cultural ecosystem. It recommends what we should see, listen to, read, and buy. It determines how many people will see our shared content. It helps us make aesthetic decisions when we create media. In professional cultural production, AI has already been adapted to produce movie trailers, music albums, fashion items, product and web designs, architecture, etc. In this short book, Lev Manovich offers a systematic framework to help us think about cultural uses of AI today and in the future. He challenges existing ideas and gives us new concepts for understanding media, design, and aesthetics in the AI era.

“[The Analytical Engine] might act upon other things besides number...supposing, for instance, that the fundamental relations of pitched sounds in the science of harmony and of musical composition were susceptible of such expression and adaptations, the engine might compose elaborate and scientific pieces of music of any degree of complexity or extent.” - Ada Lovelace, 1842.

Writing in 1842, Ada Lovelace imagines that in the future, Babbage’s Analytical Engine (general purpose program-mable computer) will be able to create complex music. In the 2017 novel by famous Russian writer Victor Pelevin, set in the late 21th century, the narrator is an algorithm solving crimes and writing novels about them. Today we exist somewhere between these two visions of cultural AI (Artificial Intelligence). Algorithms are frequently used to write music, but they don’t really “understand” the human world and human meanings. Whether the latter will ever happen is unclear.

The original vision of AI was about automation of cog-nition. Today, AI also plays a crucial role in culture, in-creasingly influencing our choices, behaviours, and imaginations. For example, it is used to recommend photos, videos, music, and other media. AI is also used to suggest people we should follow on social networks, to automatically beautify selfies and edit user photos to fit the norms of “good” photography, and to generate and control characters in computer games. While algorithms have been employed in artistic creation by artists since the 1960s, today industrial scale “cultural AI” is built into devices and services used by billions of people. Instead of being an instrument of a single artistic imagination, AI has become a mechanism for influencing the imaginations of billions. Gathered and aggregated data about the cultural behaviours of multitudes is used to model our “aesthetic self,” predicting our future aesthetic decisions and likes – and potentially guiding us towards choices preferred by the majority. The integration of AI into the everyday cultural lives of billions of people raises important questions about the future of culture, aesthetics, and taste. In this short book, I discuss some of these questions.

THE MEANING OF “ARTIFICIAL INTELLIGENCE”

In the original vision of AI in the 1950s-60s, the goal was to teach a computer to perform a range of cognitive tasks. In this vision, a computer would simulate many operations of a single human mind. They included playing chess, solving mathematical problems, understanding written and spoken language, and recognizing the content of images. Sixty years later, AI became a key instrument of modern economies, deployed to make them more effi-cient, secure, and predictable by automatically analyzing medical images, making decisions on consumer loans, filtering job applications, detecting fraud, and so on. AI is also seen as an enhancer of our everyday lives, saving us time and effort. A good example of this is the use of voice interface instead of typing.

But what exactly is “Artificial Intelligence” today? Besides original tasks that defined AI such as playing chess, recognizing objects in a photo, or translating between languages, computers today perform endless “intelligent” operations. For example, your phone keyboard grad-ually adapts to your typing style. Your phone may also monitor your usage of apps and adjust their work in the background to save battery. Your map app automatically calculates the fastest route, taking into account traffic conditions. There are thousands of intelligent, but not very glamorous, operations at work in phones, computers, web servers, and other parts of the IT universe. Therefore, in one sense, AI is now everywhere. While some AI roles attract our attention – such as Google’s Smart Reply function that suggests automated email replies (used for 10% of all answers in Google’s Inbox app in 2017) – many others operate in the gray everyday of the digital society. Why are some intelligent tasks that computers can accomplish seen as “real” AI, and others are not?

Observers and historians of the AI field talk about “AI ef-fect.” It means that “when we know how a machine does something ‘intelligent,’ it ceases to be regarded as intelli-gent” (Promise of AI Not So Bright, 2006). That is, after the AI field solves a problem and the solution is implemented in the industry, it is no longer seen as part of the field. Paradoxically, we tend to only see the challenging and not-yet-solved problems as belonging to Artificial Intelli-gence – and this creates the impression that AI research has not been successful throughout its long history. The dramatic increase in computer capacities, the ubiquity of digital devices and networks, and the challenges and opportunities brought by the “big data” trend of the 2000s have also affected AI. We moved from automation of a single mind to a kind of “super-cognition.” Think, for ex-ample, of search engines such as Baidu, Yandex, Bing, and Google that continuously scan the web and index billions of websites and blogs. When you enter a search query, a search engine instantly returns relevant results drawn from such an index. No single human could ever perform such a feat. The scale of digital culture demands intelligence that is similar to a human qualit tively but operates on a quantitatively different scale.

AESTHETIC AI

As I already pointed out, the original vision of AI was about automation of cognition. Despite the difference in scale, super-cognition still follows this paradigm. So, when people talk about the great successes of AI in re-cent years, the examples used are the same task defined at the field’s start many decades earlier: natural speech understanding, automated translation, and recognition of

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Trevor
Paglen
& Kate
Crawford,
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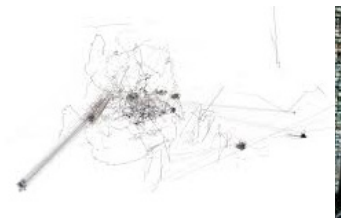
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Ben
Grosser,
Com-
puters
Watching
Movies
(2013–
2018)

objects in photos. But what is perhaps less obvious is that AI now plays an equally important role in our cultural lives and behaviours, increasingly automating the processes of aesthetic creation and aesthetic choices. Consider, for example, these selected examples of AI adoption in just one single cultural field – digital photography. I divided them into two categories: assistance in selecting appropriate images from large (often massive) collections, and assistance in the creation/editing of new content. (Note that AI can assist the human selection process or be employed for completely automatic selection.)

Selecting from existing content:

1. Image sharing services and marketplaces use AI to predict the content of images and assign keywords to them (Grigonis, 2016).

2. Instagram’s Explore screen recommends images and videos to each user based on a combination of many factors (not only based on what the user liked in the past).

3. Yelp automatically selects the best photos to illustrate the numerous businesses listed.

4. The Roll app from EyeEm automatically rates the aesthetic quality of user photos, while EyeEm image marketplace assigns such scores to submitted photos.

5. Huawei ran a photo contest where submitted photos were judged by AI: “Trained using 4,000,000 images taken by professional photographers and picture editors, the AI will then give each photo a personalized AI score based on parameters such as focus, jitter, deflection, color, and composition” (Hillen, 2018).

Creation / Editing of new content

1. Photo apps can automatically modify catured photos according to the norms of “good photography.”

2. Other apps “beautify” selfies and portraits. Tencent’s (leading Chinese IT company) long list of capabilities in this area demonstrates the range of automated adjustments possible: “dermabrasion, skin whitening, eye enlargement, face thinning, removing acne, adding eyelids, changing skin color, recognizing face color and applying foundation, applying lip gloss, eyebrow shaping, and applying other makeup” (Tencent YoTu Lab, 2018).

3. Photoshop 9.1 uses AI in its “subject select” function to automatically select objects from a background. 4. Phone cameras analyze the 3D layout of a scene and blur backgrounds in portraits and selfies (Associated Press, 2016).

5. The Huawei Mate 10 phone camera (released 10/2017) uses AI to analyze what it sees. It then classifies it into one of several scene types and selects the appropriate parameters for capturing a given scene – even before you decide to take a photo.

AI AND AESTHETIC DIVERSITY

One important trend we can see in the examples above is a movement towards gradual automation (semi or full) of aesthetic decisions – recommendation engines suggesting what we should watch, listen to, read, write, or wear; devices and services that automatically adjust the aesthetics of captured media to fit certain criteria; software that rates the aesthetic quality of our photos, etc. This development raises big questions about the future of culture. Does such automation lead to a decrease in aesthetic diversity over time? Is this inevitable, or are there other forces that may counteract this, increasing diversity? To illustrate what this means for image culture, the question may be rephrased, for example, as following: Do automated enhancements and edits that mobile cameras and photo sharing services apply to user photos make them less diverse aesthetically? Will further AI integration in user photography devices and image sharing sites lead to standardization of “photo imagination?” Do search and recommendation engines or functions such as Instagram’s Explore tend to show the same images (or many variations of images with certain content, or perhaps only images with a certain “professional” aesthetic) to lots of people, thus diminishing the diversity of what we see? But AI, algorithms, and user interfaces of digital services, apps, and products may also be increasing aesthetic diversity. For example, digital cameras and photo apps have many functions for customization. On my camera (Fuji E-3), I can choose the shutter speed, aperture, ISO, desired levels of highlight and shadow tone, color density, sharpening, grain, dynamic range, noise reduction, and film simulation filters (Fujifilm Corporation, 2017). Free photo editing apps such as Snapseed, which are used by many people to prepare their photos for Instagram, also offer a large number of editing tools comparable to professional desktop software such as Photoshop and Lightroom. Over time, phone cameras and photo editing apps have been adding more and more controls, and a lot of them are now free. Therefore, while the gradual AI integration into phone cameras and sharing sites may contribute to a decrease in aesthetic diversity, the simultaneous addition of more and more controls to cameras and photo apps may have the opposite effect. Or consider recommendation engines. They can be programmed to recommend items that are already the most popular among other users, thus decreasing your chances of



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infinite
duration,
2015.

seeing more varied items. Alternatively, they can be programmed to expose users to more diverse items including ones that they most likely would not find on their own. A person may use many engines for different media daily, and they may all be programmed differently, so it would be unwise to assume that all engines together push users in any one particular direction. And since any industry engine uses many different inputs to come up with its recommendations, the items it recommends to a user may both expose her to what is already popular, and to what she would not find on her own. In one 2010 quantitative study, researchers set out to “investigate the impact of YouTube’s recommendation system on view diversity to understand whether the recommendation system helps users to discover videos of interest, but not necessarily popular [ones], or is more likely to recommend popular videos only.” Their conclusion was “that the current recommendation system helps to increase the diversity of video views in aggregation” (Zhou, Gao, & Khemmarat, 2010) – but this can be different in different times, as YouTube changes its algorithms.

Of course, a number of other trends also influence aesthetic diversity in contemporary culture besides computational technologies. The rise of the world wide web and social networks, growth of international travel, globalization of consumer economies and advertising, zero cost telecommunication, growth of foreign student enrollment, growth of remote work, and the rise of Japan, followed by Korea and then China, as exporters of cultural products and images are just some examples among many other developments all playing a role. On the one hand, they are making the world into a single global village – or if you like, a single cultural marketplace, where certain images, ideas, values, narratives, products, and styles are marketed to everybody and available everywhere, and this may decrease diversity. On the other hand, the same trends may also be increasing diversity because local cultural DNAs become available globally.

Given that many developments are influencing global aesthetic diversity, the role played by cultural AI is probably not the most significant yet – but it is likely to grow in the future for at least two reasons. First, billions of people who still don’t have access to internet and smartphones will get this access and start using the same AI-driven recommendation engines, automated aesthetic editing of captured media, selfie-beautifying apps, and so on. Second, the automation of aesthetic decisions we have seen so far is still at an early stage, with many more things to come. For example, right now people take photos themselves, cameras apply some aesthetic adjustments at the time of capture, and then a person may use editing software to do further adjustments. But it is easy to imagine a future scenario where cameras will themselves choose what

and when to capture to give us the most satisfying photos that fit certain concepts and aesthetic ideals. In fact, the Google Clips video camera released in January 2018 is already doing this. The camera is fully AI-based. It uses computer vision to recognize people and pets and certain emotional expressions, and was trained by professional photographers to make “good” videos with proper composition, interesting actions, etc. (Lovejoy, 2018).

MESURING DIVERSITY

So how can we know if aesthetic diversity in contemporary culture – or even only in one area such as photography – is growing or decreasing? Maybe we can use AI itself to start answering such questions more precisely, as opposed to guessing or just following our intuitions that are often wrong?

Since the middle of the 2000s, hundreds of thousands of computer and social scientists have been quantitatively analysing massive samples of contemporary digital culture, including billions of posts and user interactions on Facebook, Twitter, Instagram, Flickr, Pinterest, and other social networks, as well as on recommendation sites such as Yelp, creatives networks such as Behance, etc. (I will describe relevant examples of such research below). They developed many quantitative measures that describe – or attempt to – some aspects of culture such as structures of sharing in social networks or uniqueness and originality of user-created images. If we think within this paradigm, we can also propose measures of aesthetic diversity, and apply them to some cultural areas and types of media. Since we can often access user content shared online in the past (for example, content shared on Flickr since 2004, or Instagram since 2010), we can also calculate how diversity in some cultural areas is changing over time. This would require developing formal measures of aesthetic diversity for different media and cultural fields, from fashion and interior design to cinema and music. And this would be very useful by itself because it would allow us to look at contemporary culture in new ways. Although such formal definitions will never fully account for our aesthetic experiences in many cases, they can still help us by providing new concepts for thinking about global digital culture.

Note that we would need to differentiate between different types of diversity. One is diversity of content, i.e. the items being created in a particular cultural area. For example, in the case of photography, this diversity includes types of subjects, techniques, and styles in photographs. Another is diversity in users’ choices, similar to how it was analysed in the YouTube study previously cited. For example, contemporary fashion designers around the world may be

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Mario
Klingemann,
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Identity
(2018)



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Machine
Hallucinations”,
Refik
Anadol.
2019



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Jake
Elwes,
The Zizi
Project
(2019)

creating items that can have very diverse styles, silhouettes, forms, volumes, materials, textures, and colours, but the diversity of items being purchased and worn by people worldwide may be much smaller. Or, it can be much bigger since today many people mix different items in their outfits, creating composite looks that designers and retailers do not offer. Other types of diversity can also be defined as appropriate. The idea of measuring aesthetic diversity in global contemporary culture allows us to make other interesting distinctions. One is between global and local diversity. If we measure a sufficient number of items globally, culture in many local places may look very homogeneous in comparison to the full range of choices available worldwide. But, if we zoom into such places, what looked like small hills will now look like mountains, so to speak – i.e., we will realize that these places are quite diverse if viewed on their own.

Related to this distinction is another one – between objective and subjective measurements. (We can also call this analyst vs. users’ perspectives.) So far, we assumed that our measurements are taken from an abstract, all-seeing point of view. All items or user choices are placed on a single scale and viewed and measured from the outside. Such perspective is standard in biology when we are measuring Earth’s biodiversity – for example, counting the number of distinct species in a given habitat, or on Earth as a whole. But in the case of culture, we may alternatively consider how diversity and differences between items inside a habitat is perceived by members of this habitat itself. In this perspective, local traditions and conventions determine whether some items or choices are perceived as radical or not – and not their objective characteristics alone. To continue with fashion examples, in many Western cities people commonly wear many colours, and this is not seen as radical. But in a city like Seoul where the palettes of greys, whites, and blacks dominate how people dress, the appearance of outfits with very saturated colours, and especially more than one such colour together, will be noticed and perceived as being outside the norm.

LIMITS TO AUTOMATION

Will AI replace professional cultural creators – media, industrial, and fashion designers, photographers and cinematographers, architects, urban planners, and so on? Will countries and cities worldwide compete as to who can more quickly and better automate their creative industries? Will countries and cities (or separate companies) that figure out how best to combine AI and human skills and talents get ahead of the others?

Today AI gives us the option to automate our aesthetic choices (via recommendation engines), assists in cer-

tain areas of aesthetic production such as consumer photography, and automates other cultural experiences (for example, automatically selecting ads we see online). But in the future, it will play a larger part in professional cultural production. Its use of helping to design fashion items, logos, music, TV commercials, and works in other areas of culture is already growing. But currently, human experts usually make the final decisions or do actual production based on ideas and media generated by AI. The well-known example of Game of Thrones (American fantasy television drama series that premiered in 2011) is case in point. The computer suggested plot ideas, but the actual writing and the show’s development was done by humans. We can only talk about fully Aldriven culture where AI will be allowed to create the finished design and media from beginning to end. In this future, humans will not be deciding if these products should be shown to audiences; they will just trust that AI systems know best – the way AI is already fully trusted to choose when and where to show particular ads, as well as who should see them.

We are not there yet. For example, in 2016 IBM Watson created the first “AI-made movie trailer” for the feature film Morgan (Mix, 2016). However, AI only chose various shots from the completed movie that it “thought” were suitable to include in the trailer, and a human editor did the final selection and editing. In another example, to create a system that would automatically suggest suitable answers to the emails users receive, Google workers first created a dataset of all such answers manually. AI chooses what answers to suggest in each possible case, but it does not generate them. (The head of Google’s AI in New York explained that even one bad mistake in such a scenario could generate bad press for the company, so Google could not risk having AI come up with suggested answer sentences and phrases on its own.)

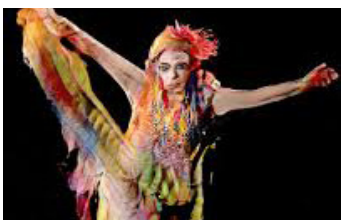
It is logical to think that any area of cultural production which either follows explicit rules or has systematic patterns can be in principle automated. Thus, many commercial cultural areas such as TV dramas, romance novels, professional photography, music video, news stories, website and graphic design, and residential architecture are suitable for automation. For example, we can teach computers to write TV drama scripts, do food photography, or compose news stories in many genres (so far, AI systems are only used to automatically compose sports and business stories). So rather than asking if any such area will be automated one day or not, we need to assume that it will happen and only ask “when.”

This sounds logical, but the reality is not so simple. Starting in the 1960s, artists, composers, and architects used algorithms to generate images, animations, music, and 3D designs (“Computer Art,” n.d.). Some of these works

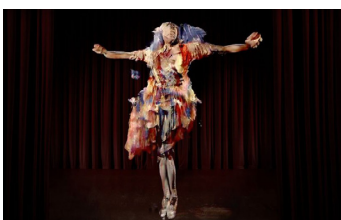
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Jake
Elwes,
The Zizi
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(2019)

have entered the cultural canons. They display wonderful aesthetic inventiveness and refinement. However, in most cases they are abstract compositions with interesting and complex patterns, but without direct references to the human world. Think of such classics as abstract geometric images by Manfred Mohr (1969- 1973) (Mohr, n.d.), John Whitney's computer animation Arabesque (1975), or Iannis Xenakis's musical compositions Atrées and Morsima-Amorsima (1962) (Maurer, 1999). There is no figuration in these algorithmically generated works, no characters like in novels, and no shots of the real world edited together into narratives like in feature films.

Now compare these abstract algorithmic classics with current attempts to automatically synthesize works that are about human beings, their worlds, their interests, emotions and meanings. For example, today Google Photos and Facebook offer users automatically created slideshows and videos edited from their photos. The results are sometimes entertaining, and sometimes useful, but they can't be yet compared to professionally created media. The same applies to images generated by Google engineers using DeepDream neural net (2015-) and later by others who used the same technology (DeepDream, n.d.). These AI creations in my view are more successful than the automatically generated slideshows of user photos, but this is not because DeepDream is a better AI. The reason is that 20th century visual art styles tolerate more randomness and less precision than, for example, a photo narrative about a trip that has distinct conventions and restrictions on what and can be included and when. Thus, in the case of DeepDream, AI can create artistically plausible images which do refer to the human world because we consider it "modern art" and expect big variability. But in the case of automatically edited slideshows, we immediately know that the computer does not really understand what it is selecting and editing together.

AI AND GENERATIVE CONVENTIONS

Creating aesthetically satisfying and semantically plausible media artefacts about human beings and their world may only become possible after sufficient progress in "artificial general intelligence" (AGI, also referred to as "strong AI") is made. In other words, a computer would need to have approximately the same knowledge of the world as an adult human. However, this is not only a matter of progress in AI research. Whether an algorithmic creation looks plausible or not also depends on genre conventions. In some cases, even very simple AI can produce satisfying results. In 2002-2005, I collaborated with Andreas Kratky on Soft Cinema – a semi-automated system for the making of procedural narrative films (Manovich & Kratky, 2005; Manovich, 2005). Using software we developed, we

produced three such films. Each film used a database of a few hundred short video shots we recorded. The choice of shots for the films and their editing in time was done by software using parameters we selected.

The project was shown as an installation in 45 exhibitions. During the duration of a singular exhibition, which sometimes would last a few months, the software was continually editing a film in real-time by selecting short video clips from a database. The complete narrative for each film lasted between 7 and 13 minutes, depending on the film. After one version played, a computer immediately started editing and showing the next version. Rather than trying to simulate conventions of mainstream narrative cinema or documentary filmmaking, we instead took as our inspiration experimental 20th century films. Specifically, we used the principle that narration and visuals do not always have to correspond. As a voice-over narrated a story, a computer program was selecting short video clips from a database and editing them together using metadata and rules we established. The program was also generating screen layouts which used a Mondrian-like grid to display anywhere from one to six videos in the same frame. The sequences of clips did not directly illustrate the narrative. However, since all clips in each film database shared certain semantic and visual references, the overall result of this semi-automated process looked meaningful. The mind of a viewer created connections between the narrative content and visuals of the clips playing on the screen. Thus, the more "loose" and associative conventions of "avant-garde" or "experimental" cinema turned out be much easier to simulate than a conventional narrative film. The latter requires much tighter coordination between all shots. The viewer would immediately notice all mistakes AI made, while in our Soft Cinema mistakes were no possible in principle, since the selection and screen placement of clips did not directly illustrate the narrative. However, it is likely that we will also gradually learn to use AI to generate works in cultural genres which have a lot of rules and constraints. For example, synthesized 3D human characters and conversational agents are gradually becoming more realistic. This is a very challenging area because human verbal communication is certainly a very strict "genre" – if a speaker makes a gesture or facial expression that does not correspond to what he or she is saying, we will notice this immediately. The use of algorithms in design and architecture (often called "parametric design") is another mature area. Perhaps one day we can delegate to computers the creation of all details of an entire city, from planning to architecture, landscaping, traffic management, and all infrastructure. But when this happens, will we be able to prevent such an AI-metropolis from imposing on, regulating, and rationalizing our existence in the name of progress and human happiness – like the urban visions of the 17th – 19th centuries, Eu-



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Jerome Saint-Clair, Predictive Art Robot (2015–2018)

ropean utopias of ideal human communities, Le Corbusier’s 1920s radical urban rationalization proposals, and cybernetic and mechanical visions in Godard’s Alphaville and Tati’s Playtime?

If all creative and knowledge work will become the domain of AI, what will be left for humans? What will be the purpose of our existence? Watching endless films created by AI, listening to AI-generated music, and being driven in driverless cars around AI-generated cities? Many modern thinkers and artists have envisioned a future where humans, liberated by machines from mechanical and boring work, will be engaging only in play and art (e.g., Constant’s New Babylon). But if automation of cultural production by AI continues, eventually it will be these AI playing and making art – not us.

AI AS A CULTURE TERORIST

Automation of cultural production may use AI based on a system of explicit rules, or it may use a different approach called “supervised machine learning.” Advances in “deep learning” (particular methods for supervised machine learning) in the 2010s have made the latter approach very popular today. In one common cultural application of supervised learning, a computer is first fed many examples of works of particular genres, styles, and media, and it gradually learns the patterns common to each genre or style. After that, a computer is given new works it has not seen before and uses the patterns it learned to classify those works. For example, in the case of creating the movie trailer for Morgan, a computer was given 100 horror movies. In these movies, “each scene was tagged with an emotion from a broad bank of 24 different emotions and labels from across 22,000 scene categories, such as eerie, frightening, and loving” (Smith, 2016). The computer learned “the types of scenes that categorically fit into the structure of a suspense/horror movie trailer.” So, after it was given Morgan, it selected the scenes that it believed to be the best ones for the trailer.

We can say that AI, implemented in this way, acts as a cinema (or art, video game, fashion, etc.) theorist and historian. These researchers also study many works created in particular places and historical periods to find common patterns. Their findings become part of the history and theory of this area.

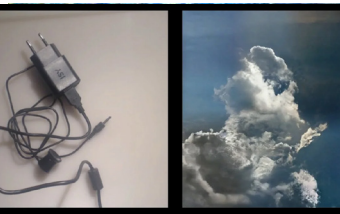
However, this is a crucial difference between an “AI culture theorist” and a human theorist/ historian. The latter comes up with explicit principles that describe how a cultural area function. For example, the standard textbook used in universities in numerous film studies courses – The Classical Hollywood Cinema – provides answers to questions such

as “How does the typical Hollywood film use techniques and storytelling forms of the film medium?” (Bordwell, Staiger, & Thompson, 2010) But at present, the frequent result of training AI ((i.e., use of deep neural networks) on many cultural examples is a black box. Given new examples, it can classify them correctly – for example, it can decide whether a particular film belongs to “classical Hollywood cinema” or not. But we often don’t know how a neural network came up with this decision. Similarly, a neural net can be trained to distinguish between works of different artists, fashion designers, or film directors. And it can also generate new objects in the same style. But often we don’t know what exactly the computer has learned. (However, many computer scientists are working to develop methods to make black boxes created by neural networks more transparent and for it to be possible to “audit” them.)

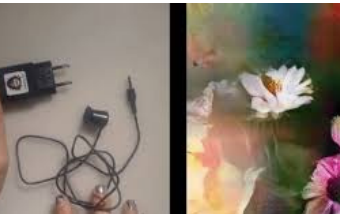
This is one of the key issues surrounding cultural uses of AI. Are the results of machine learning interpretable, or are they only a black box which is efficient in production but useless for human understanding of the domain? Will the expanding use of machine learning to create new cultural objects make explicit the patterns in many existing cultural fields that we may not be aware of? And if it does, will it be in a form that will be understandable for people without degrees in computer science? Will the companies deploying machine learning to generate movies, ads, designs, images, music, urban designs, etc. expose what their systems have learned?



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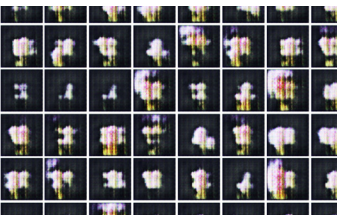


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Memo
Akten,
Learning
to See
Series
(2017–)



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Anna
Ridler,
Mosaic
Virus
(2019)

Hito Steyerl (2022)

[D] Crespo, S. (n.d.). Neural Zoo. Retrieved December 14, 2025

Animal Spirits

Animal Spirits is a video installation by Hito Steyerl, consisting of a one-channel HD video film and a live computer-generated animation as well as spatial installation elements. It premiered in the artist's solo exhibition A Sea of Data at the Museum of Modern and Contemporary Art, Seoul, 2022. The live-generated animation is changing according to data collected from sensors. The continuously changing animation follows a light-sparkling cloud of particles, travelling through what looks like a natural cave.

The narrative video begins with an introduction of the term 'Animal Spirits': Coined by the economist John Maynard Keynes, Animal Spirits describes human emotions, urges and instincts that may drive economic markets out of control or ordinary development. The film interlaces historical film footage with animation, motion graphics and documentary interviews. It touches on topics of post-pandemic conditions of cultural production, the popularity of cryptocurrencies and NFT-works, and possibilities for (farming and fermentation inspired) systems of decentralization and autonomy.[D]

Generative Art

“Generative ai art as
candy crush”

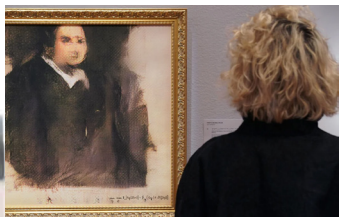
Zylinska, J. (2020). Generative AI Art as Candy Crush.
In *AI Art: Machine Visions and Warped Dreams*. Open
Humanities Press.



The 2017 edition of ars electronica took place under the banner of ‘ai: artificial intelligence / das andere ich’. in the catalogue accompanying the event, the ars director Gerfried stocker gushed in a tone that could have come straight from silicon Valley investor publicity: ‘the latest developments in artificial intelligence are truly astonishing, and they will soon be advancing exponentially. never before has so much investment capital been [put] in pursuit of successful technologies and promising innovations’ (stocker 2017, 16). it is thus perhaps not accidental that much of the artistic research around ai is facilitated and sponsored by the main players within what has become known as platform capitalism: Google, amazon, Facebook and apple. Much of ai art is precisely platform art: generating visual and algorithmic variations within the enclosed system while teasing the public with the promise of novelty.



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Mario
Klingemann,
Neural
Glitch /
Mistaken
Identity
(2018)

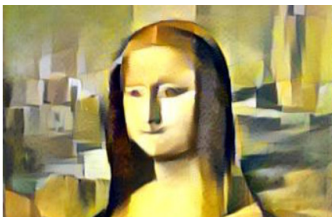
Unlike most of the installation based robotic art discussed earlier, this kind of ai art, described with the term ‘generative’, takes place across computer networks. Kindly put, much of generative ai art celebrates the technological novelty of computer vision, fast processing power and connection-making algorithms by regaling us with a dazzling spectacle of colours and contrasts as well as the sheer volume of data. Unkindly put, it becomes a glorified version of candy crush that seductively maims our bodies and brains into submission and acquiescence. art that draws on deep learning and big data sets to get computers to do something supposedly interesting with images often ends up offering a mere psychedelic sea of squiggles, giggles and not very much in-between. It really is art as spectacle. It is with this critical mindset, tinged with curiosity and fascination, that i approach much of the current work based on neural networks, code and algorithms by artists such as Gene Kogan, Mike Tyka, Memo akten and Mario Klingemann. Kogan’s Neural Synthesis was developed for the ‘creativity exhibition’ at nips (neural information processing systems) conference in 2017. The 2’41” video offers a psychedelic transmogrification of garishly multicoloured figures, faces and abstract patterns, slowly emerging before the viewer’s eyes. Kogan’s technique involves repeatedly optimising pixels in an image ‘to achieve some desired state of activations in a convolutional neural network’ (Kogan and Ikkchung 2017) – i.e. a deep learning network, consisting of many layers of artificial neurons, which assumes the data it is being fed consists of images. That ‘desired state’ may end up looking like a dalmatian, a starfish or, indeed, a human face. The final layer in the network ‘essentially makes a decision on what the image shows’ (Mordvintsev 2015). an advanced version of this technique gained recognition in 2015, when Google released it to the public under the name of deepdream. put simply, deepdream works by identifying and enhancing patterns in images, leading to the algorithm ‘finding’ human eyes or puppies in any regular photographs.

The programme was originally developed ‘to help scientists and engineers to see what a deep neural network is seeing when it is looking in a given image’ but was quickly repurposed as a creative tool, which subsequently underwent various iterations. Works produced using the deepdream algorithm were described with the term ‘inceptionism’, ‘which derives its name from the network in network paper by Lin et al. [12] in conjunction with the famous “we need to go deeper” internet meme’ (szegedy et al. 2014). Yet artists as well as the wider public quickly got bored with what a Wired journalist has described as ‘Google’s trippy neural network, which chews up reality and spits out slugs, dogs and eyes’ (Temperton 2015).

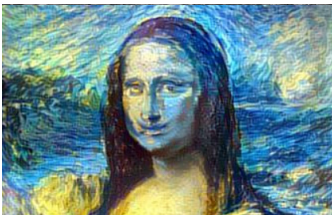
A Google scientist called Mike Tyka, who was the author of some of the first large-scale artworks made with deepdream, took some further steps with generative ai. his Portraits of Imaginary People, featured at ars 2017 and produced by generative neural nets, are photorealistic yet also somewhat ‘oil-painty’ images of humans of different sexes, ethnicities and ages. To create them, the artist fed thousands of photos of faces from Flickr to a machine-learning programme called a generative adversarial network (Gan). Gans use two neural networks, where a neural network is basically an algorithm that is designed from bottom up, in a way that (supposedly) imitates the way the human brain works. The two neural networks in a Gan are placed in an adversarial relationship, with one tasked with generating convincing and correct input, the other – with controlling and improving upon this input, according to the truth/falsehood criteria. Their ongoing interaction makes both networks improve with time, learning from each other while trying to outdo each other in obtaining ‘good’ results. requiring more knowledge of programming than the deepdream interface did, Gans are now a frequent tool in the arsenal of many ai artists, especially those with science and engineering background. Given the source material used to train machine vision, it is perhaps unsurprising that canvas for such experiments is frequently provided by human faces. in a similar vein to Tyka’s experiments, Memo akten also explores human portraiture. his Learning to See: Hello, World! is a ‘deep neural network opening its eyes for the first time, and trying to understand what it sees’ (akten 2017). This is a network that has not yet been fed anything. By being presented with a scanned image, via a surveillance camera, of a person standing in front of its computer host, the network attempts to figure out what it is seeing by identifying patterns in the information it has received. When faced with too much information, the network, like its human counterpart, also supposedly ‘forgets’. as the artistic output, the human viewers are presented with a moving set of transmogrified, somewhat painterly faces.

But it is the work of Mario Klingemann – a self-proclaimed ‘skeptic with a curious mind’ working with neural networks, code and algorithms – that is worth looking at in more detail, partly due to the attention it has received but also because there is something really intriguing happening within (and around) his images. another exhibitor at ars electronica 2017, he has also been an artist-in-residence at Google arts & culture: an online platform run by the silicon Valley giant and featuring curated collections of artworks and other cultural artefacts. Klingemann’s work builds on open source code written by engineers and shared with others who want to experiment with various ai applications, which he then tweaks for his own artistic purposes. Feeding large sets of data drawn from publicly available digital collections such as that of the British Li-

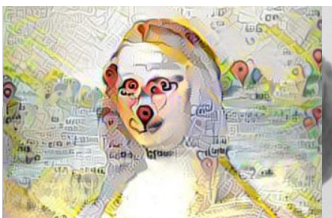
[102]
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Gene
Kogan,
Style-
transfer.
art



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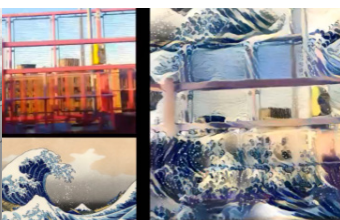


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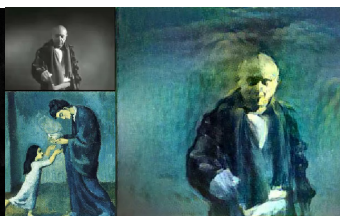


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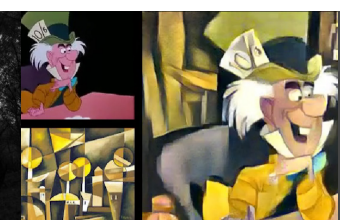
[105]
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Gene
Kogan,
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brary or the internet archive into a highpower computer, Klingemann aims to establish unusual encounters and connections between images and data points, while also visually rendering them in an interesting way – with the human artist ultimately deciding on this interestingness or its lack. his goal is ‘to create algorithms that are able to surprise and to show almost autonomous creative behaviour’ (Klingemann) with that sheepish ‘almost’ obscuring what is ultimately a largely mechanical process, although performed at a speed and intensity that exceeds that of any human reader of data. as a result, we get projects such as Superficial Beauty, featuring portraits that have been generated and enhanced by generative adversarial neural networks (Gans), or Neural Face, a collection of images of human faces scraped from various online repositories and programmed to ‘evolve’ into new faces. Using the neural network that provides biometric data for what looks like a human face, the artist trains the neural network in making new faces – which end up looking more photorealistic if he trains the network on photos, or more ‘artistic’ if he uses engravings or paintings as his training set. he also experiments with a less linear approach in that he not only derives faces from the face markers he feeds into the neural network but also derives face markers from pictures of faces and, finally, cross-feeds the data and the ‘noise’ between the two models to see what happens. Klingemann then combines the results obtained into videos showing multiple featureless faces seamlessly and dreamily morphing into one another (although with still relatively recognisable features of pretty white young girls), with an occasional exacerbation of a feature such the eye or the lip appearing slightly out of place.

Part Dali, part manga, part screensaver art, these rather kitsch images produced by the likes of Tyka, Kogan, Akten and Klingemann, especially in their moving variety, attempt to seduce viewers with a mildly fascinating transformation of humanist representationalism under the banner of ‘uhm... what now?’. There is thus little difference, both aesthetically and in terms of creativity, between Google’s deepdream, which comes up with surreal images by finding and enhancing patterns between various data points, and what Klingemann terms ‘cameraless neurophotography’, which produces mesmerising yet somewhat creepy images made from parts of other images. Turning the mindless generation of images into an art form, the artist remains seemingly unaware that his art serves as a blanket for technical entrepreneurs’ rather more pernicious, although not any less surreal, deep dreams. Yet, as hito steyerl argues in Duty Free Art, ‘these entities are far from mere hallucinations. if they are dreams, those dreams can be interpreted as condensations or displacements of the current technological disposition. They reveal the networked operations of computational image creation, certain presets of machinic vision, its hardwired ideologies and preferences’ (2017).

Indeed, generative art of this kind is premised on the banality of looking, with perception understood as visual consumption and art reduced to mild bemusement. it therefore does something more pernicious than merely introduce a new ‘new aesthetic’: slightly uncanny, already boring. ‘[h]ardwiring truly terrifying jargons of cutesy into the means of production’, it enacts what steyerl terms ‘a version of corporate animism in which commodities are not only fetishes but morph into franchised chimeras’ (steyerl 2017). in projects of this kind, the artist’s goals are clearly more aligned with the current developmental trajectories of corporately-funded ai, even if naturalised via the language of evolution.

The language of evolution features prominently in the gen-art boys’ discourse but theirs is a rather normative and linear idea of evolution understood as linear progression on an upward trajectory, not a process of false starts, zigzags and pointless repetitions. it thus fits perfectly with the technoevolutionary narrative of silicon Valley ‘visionaries’ and investors, who are all sharpening their teeth at the next supposed technological leap. Through both their art and their discourse on art, Kogan, Tyka, akten and Klingemann adopt a worryingly uncritical instrumentalism, where seemingly child-like curiosity is underpinned by the progressivist model of technological expansion towards some kind of improvement – of accuracy, data sets and, ultimately, art as we know it. They are thus poster boys for ai art as underwritten by Google in the way they incarnate the very ideas of progress, innovation and upward trajectory that drive the gung-ho progressivism of the ai 2.0 era. in their experiments that derive art from inputting data and noise into neural networks’ feedback loops, they seem unwilling, and unable, to pursue any of the more serious questions raised by steyerl in her indictment of computer vision art: ‘Which faces appear on which screens, and why? ... Who is “signal,” and who disposable “noise”?’ (steyerl 2017). steyerl highlights the fundamentally political dimension of separating signal and noise in any kind of data set, with pattern recognition resonating ‘with the wider question of political recognition’ (2017)

Dreamy neural network art of this kind thus ultimately has a pacifying effect, anaesthetising us into the perception of banal sameness that creates an illusion of diversification without being able to account for differences that matter – for why and where they matter, when and to whom. it thus ends up enforcing a mode of existence that philosopher Franco ‘Bifo’ Berardi has called a ‘neurototalitarianism’. neurototalitarianism manifests itself in the utter intensification of semiotic (but we may also add, visual) simulation, in the end resulting in the sensation of panic in the social – and individual – neuro-system. ‘in this condition of panic, reason becomes unable to master the flow of events or to process the semio-stimulations released into

the infosphere’ (Berardi 2017). Would it be a step too far to state that ai art can actually be mobilised, wittingly or unwittingly, in the service of neurototalitarianism, using cutsey kittens and spaghetti with eyes to capture not just the attention but also the cognitive and neurological sphere of the modern political subject? This form of art enforces what Berardi has described as ‘mental subsumption’ (2017), whereby the automation of vision and of cognitive activity paves the way for the emergence of constantly stimulated yet passive subjectivity. it thus ultimately works in the service of neoliberalism, a mutated form of capitalism that, according to Byung-chul han, ‘has discovered the psyche as a productive force’. Yet what gets produced here is ‘mental optimisation’ (han 2017), for which Big data, algorithms and ai serve as perfect conduits. Han calls this state of events a psychopolitics, which is his variation on Foucault’s notion of biopolitics, whereby the subject mobilises technologies of the self not to create any forms of freedom but rather to succumb, ‘willingly – and even passionately’ (2017), to auto-exploitation (fig. 9). The brain, the mind, the eye, the heart, the finger, the tongue – the whole cortico-corporeal apparatus – all turn into a platform for the actualisation of this new form of psychopolitics, which drives ‘the capitalism of “Like”’ (han 2017). Vision and perception are constantly stimulated in the process with the promise of a digital caress: they are being controlled by pleasure, rather than by coercion or prohibition, as was the case with Foucault’s disciplinary model of society.

The self-quantification movement is one example where the self is constantly self-optimising, converting the principle of the care of the self into a form of labour but also into an aesthetic object in its own right. The process of self-optimisation is incessant, openended and always in danger of losing against someone with a higher score – yet it is also sustained by constant prompts of positivity: likes on Facebook or instagram, pings from one’s Fitbit, retweets. For han, ‘neoliberal psychopolitics seduces the soul; it pre-emptly it in lieu of opposing it. it carefully protocols desires, needs and wishes instead of “depatterning them”’ (han 2017).

There is little difference in this model between ai art based on deep learning and neural nets, and the wider internet aesthetics, with gifs, kittens and the constant flow of data on the Twitter stream or Facebook wall seeking ‘to please and fulfil, not to repress’ – but also being able to ‘even read desires we do not know we harbour’ (han 2017). The internet turns all of us into works of art, with our own self-aestheticisation via algorithms re-optimising our body and our prosthetic selves into a digital object of display: fully transparent, always on show. han proposes ‘idiotism’, a withdrawal of communication and veiling oneself in silence, as the only salutary response to this

state of events. Yet, rather than embrace this form of philosophical Luddism, i am more inspired by Berardi’s call for the emergence of new cognitive abilities, or a new morpho-genesis he seeks. The idea here ‘is to dismantle and reprogramme the metemachine, creating a common consciousness and a common technical platform for the cognitive workers of the world’ (Berardi 2017).

[108] [109]
[110]
Machine
Hallucina-
tions”,
Refik
Anadol
.2019

[108]



[109]



[110]



Sofia Crespo (2019-2021)

[E] Crespo, S. (n.d.). Neural Zoo. Retrieved December 14, 2022

Neural Zoo

Neural Zoo is an exploration of the ways creativity works: the recombination of known elements into novel ones. These images resemble nature, but an imagined nature that has been rearranged. Our visual cortex recognizes the textures, but the brain is simultaneously aware that those elements don't belong to any arrangement of reality that it has access to. Computer vision and machine learning could offer a bridge between us and speculative "natures" that can only be accessed through high levels of parallel computation.

Starting from the level of our known reality, we could ultimately be digitizing cognitive processes and utilizing them to feed new inputs into the biological world, which feeds back into a cycle.

Routines in an artificial neural networks become a tool for creation, one that allows for new experiences of the familiar. Can art be reduced to the remapping of data absorbed through sensory processes? [E]



Made By and For Humans?

“The Issue of Aesthetic Alignment”

Arielli, Emanuele. “Made By and For Humans? The Issue of Aesthetic Alignment. & Synthetic Data and AI Cannibalism” In Artificial Aesthetics: Generative AI, Art and Visual Media, by Lev Manovich and Emanuele Arielli, 2024.

●

“AI is, in large measure, philosophy. It is often directly concerned with instantly recognizable philosophical questions: What is mind? What is meaning? What is reasoning and rationality? What are the necessary conditions for the recognition of objects in perception? How are decisions made and justified?” Daniel Dennett, When Philosophers Encounter AI, 1988.

THE ISSUE OF AESTHETIC ALIGNMENT

One crucial consideration in this regard is the fact that AI systems, like humans, operate on internal models of the world that may not perfectly align with reality. The common assertion that AI “sometimes” hallucinates is, in fact, an understatement. AI systems are continuously generating outputs based on their trained models, and these outputs can be considered a form of constant “hallucination”. The key is that these hallucinations often correspond closely enough to reality or human expectations to be useful or convincing. The same can be said of humans, who also operate with models of the world that are imperfect and prone to errors. Our ability to interact effectively with our environment is largely due to the fine-tuning of our perceptual and cognitive systems over millions of years of evolution. Similarly, AI systems must be fine-tuned to human aesthetic sensitivity, but this process is far from straightforward. To program AI models that can attune to human aesthetic preferences, these systems must be calibrated to reflect the dynamic nature of human experience. This calibration must account for individual and cultural variations, as well as the fluidity of tastes and trends. A machine that models a human evaluator by learning from human judgment belongs to the domain we called “subject generations”.

A critical distinction emerges when we consider the sources of information that shape AI and human internal models of the world. While AI systems are typically trained on vast digital datasets, human perception and cognition are the products of millennia of evolutionary adaptation to the physical world. The brain of an infant is created with a DNA that guides the development of its nervous structures while it is in the mother’s womb. The ancestors of that individual going millions of years in the past interacted with the world through sensory perception and evolutionary selection allowed the genes that encode the more useful perception and classification process to be passed on. On the other side, the internet, on which AI-systems are trained, provides an incomplete and often misleading representation of reality. Even if the entirety of the web would be used as training data base, the real world is orders of magnitude more complex. Humans, too, when susceptible to forming their understanding of the world based on internet-derived information, may not always align with the complexities of real-world experience. As philosopher Merleau-Ponty (1908-1961) emphasized, human perception and cognition and our understanding of the world are fundamentally shaped by our physical and sensorial interactions with it. This perspective suggests that to truly align AI with human aesthetic sensitivity, we may need to move beyond purely digital training and incorporate embodied experiences. According to this perspective, perception should not be considered just

passive reception of data but an active engagement with the world, shaped by the body’s interaction with its environment. For AI to align with human aesthetic sensitivity, it would need to be similarly integrated into the world, potentially through the development of AI systems embedded in physical bodies. These embodied AI systems would interact with the world in a manner akin to humans, having not just sensory experiences but also having to deal with the human ecosystem of norms, conventions, and social dynamics.

This could also mean envisioning artificial agents endowed with drive and motivation, with integrated aesthetic goals and preferences. While this might seem far-fetched, we could speculate how this immersion could potentially lead to the development of a sense of agency and authorship within these AI entities. In the aesthetic context, this would require a functional definition of what it means for a system to strive for aesthetic pleasure and having edonic preferences, possibly rooted in the recognition of harmonious forms or mechanisms linked to aesthetically pleasing patterns, the satisfaction of predictive mechanisms²⁰, or the achievement of an optimal balance between uncertainty and familiarity. Such systems would not merely respond to stimuli but would actively seek to model the world in ways that reflect goals and desires. However, the question arises if AI should be limited to merely replicating human preferences and aesthetic sensitivity. We could argue that AI could potentially go beyond human aesthetics, creating entirely new forms of beauty that humans might not have conceived but that could fulfill machine goals and preferences. These machine-generated aesthetics could involve levels of complexity, that are not designed for human consumption.

Consider an AI system generating music. If it were aligned with average human preferences, it would produce melodies that mimic simple structures, such as the verse-chorus-bridge format commonly found in pop music. On the other hand, if the AI were to truly push beyond human preferences, it would generate compositions that had an extreme harmonic complexity, akin to some sort of avant-garde composers like Arnold Schoenberg, who developed the twelve-tone technique to further scramble harmonic expectations and are object of aesthetic consumption and appreciation for only a minority of well-trained people. Theoretically innovative music could just turn out to be unbearable to listen for people who are used to more traditional harmonic structures. The AI’s alignment would need to balance innovation with accessibility, potentially including settings to adjust the complexity of musical structures according to a listener’s taste or providing the public with opportunities to develop a corresponding appreciation for these aesthetic innovations. Alternatively, when translated into a form accessible

[111] [112]
Scott
Eaton,
Humanity
(Fall of the
Damned)
(2019)



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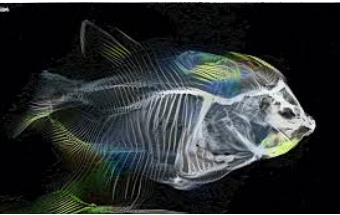


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[113]
[114]
Scott
Eaton,
Hyper-
bolic
Compo-
sitions
I & II
(2018)

[115] [116]
[117]
Sofia
Crespo,
Neural
Zoo
(2019–
2021)



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[117]

to humans these creations might be “dumbed down” and simplified to meet our perceptual and cognitive capabilities, in the same way that a chess program might have a setting that allows it to self-limit, enabling it to play in a manner that is manageable for the human player.

To make a further example, a typical novel might follow a linear plot with clear symbolic references and an AI-system which is too aligned with general human preferences might produce work that feels formulaic. However, AI might produce a text that layers multiple narratives, each with its own set of symbols and meanings, akin to James Joyce’s *Finnegans Wake*. The AI might generate dense, intertextual references that only advanced readers or literary scholars could fully appreciate. On the other hand, if it exceeds human cognitive capacities, the literature it generates could be too esoteric, requiring a dumbed down version for broader accessibility.

SYNTHETIC DATA AND “AI CANNIBALISM”

The scenarios just described, although fascinating from a speculative point of view, seem to be moving in the opposite direction to what many observers and critics appear to see in current trends. Rather than a “collaborator” with surprising capabilities assisting the artist, many foreshadow a dispossession of the creative class towards a generic homogenization of content. Rather than systems capable of opening alternative paths, AI systems would just learn from mediocre databases and tend towards a generic kitsch in his generations, stemming from the average of human preferences. To avoid a scenario in which AI runs out of high-quality data, training databases need to be carefully curated and developers need to implement strategies that continuously incorporate fresh, diverse, human-created content into training datasets. However, according to critical views, if AI-generated content endangers the careers of artists this could lead consequently to a diminished influx of new, diverse artistic styles, essential for training and improving of AI models.

Studies have suggested that, paradoxically, while generative AI may enhance individual creativity, it could also reduce the collective diversity of novel content. This phenomenon is partly due to our increasing reliance on AI systems that offer templates and pre-packaged solutions. This critique emphasizes that in the current use of generative AI, especially those systems based on prompts, the user does not work — as Michelangelo famously described — by starting with a rough block of stone and “removing” the unnecessary material to reveal the ideal form the artist has in mind. Instead, users start with a default setting of images that often features kitsch element, like idyllic landscapes, rich and colorful palettes, or

stereotypical magazine cover models that depict humans. To create works of genuine aesthetic value, the user must begin with these iconographic stereotypes—products of widely accepted standards of beauty—and must possess the capacity to achieve sufficient “escape velocity” to break away from kitsch and impose their own vision.

Moreover, the pervasive use of AI in selecting and distributing visual content may even have a feedback effect on our development of taste and preferences in the direction of uniformity. For some commentators, we are already seeing a visual aesthetic convergence, for example in the style of cafés and hotel design and in the curated looks of Instagram-inspired interiors. From this perspective, the homogenization of aesthetic experiences, amplified by AI-content, could potentially stifle creativity and diversity in visual culture. To address the issue of systems that rely too much on average preferences extracted from the training data set, researchers try to distinguish between general aesthetic assessment - the average or most

common aesthetic preferences in a given domain - and personalized aesthetic assessment.²⁴ The former refers to the analysis and modeling of average or widespread aesthetic preferences in a specific domain of interest, particularly images. The latter focuses on the analysis of data from an individual towards modeling their specific aesthetic preferences. This enables AI systems to predict a person’s rating of new content and even generate content tailored to their aesthetic preferences in a similar fashion as how algorithmic recommender system in video, music or consumer product do. The trend in personalization, on the other hand, may be also troubling. With AI systems tailored to ensure that a given profile has a completely unique content, we could move toward to a stage of hyper-personalization. This could result in aesthetic echo chambers where users would be fed with content that would just satisfy their own tastes and preferences, thus further limiting them from exposure to other aesthetics. Aesthetic products as collective and shared culture-building phenomena would be compromised by excessive individualization of the content each person would consume by means of tailored AI-generation.

Finally, another trend that is source of concern is the potential for AI-generated content to feed back into training datasets, creating a self-referential loop. Future data training sets will increasingly consist of AI-generated outputs, such as blog posts, articles, images and even fiction, as these types of content become more and more prevalent on the internet. According to recent research, this recursive loop could be disastrous for the stability of the models.²⁵ For example, AI-generated paintings, which are already trained from the history of human art, might then be included in future training datasets. The

[118] [119]
[120]
Sofia
Crespo,
Neural
Zoo
(2019–
2021)



[118]



[119]



[120]

next generation of AI, trained on this mixed dataset of human and AI generated art, would produce works that are even further removed from original human creativity. This self-referential loop has been referred to as “AI cannibalism,” and the gradual degradation over time has been dubbed “Habsburg AI” (a reference to the Habsburg dynasty, known for the recurrent inbreeding of its members), point to the fact that an “inbreeding” of data could result in a gradual loss of diversity, originality, and quality in the generated content.

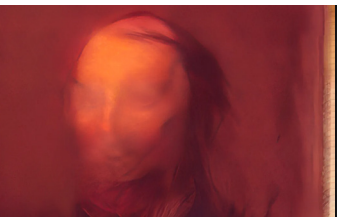
The risk is amplified by the growing use of synthetic data in machine learning training. Synthetic data refers to artificially generated information used to train AI models when real-world data is scarce, expensive, or difficult to obtain. The use of synthetic data is particularly prevalent in fields where real-world data is limited or sensitive, such as medical imaging or rare event simulation. While this approach is often necessary and beneficial in these contexts, applying similar methods to creative fields risks homogenizing the aesthetic landscape. The self-referential “cannibalism” of systems that learn from their own outputs and rely on synthetic data raises the risk of qualitative degeneration, but it could also have negative effects just from a quantitative perspective.

We can recall the case of composer David Cope, who, to overcome a creative block, began developing a system in the 1980s that could generate thousands of musical compositions in a specific style. Generative AI enormously facilitates the transition point between an imagined idea, vaguely present in our minds, and its realization. However, this ease in content generation can lead to hyper-production and an inflationary flood of content that does not necessarily translate to creative abundance or innovation. Instead, it can result in potential perceptual and cognitive overload.

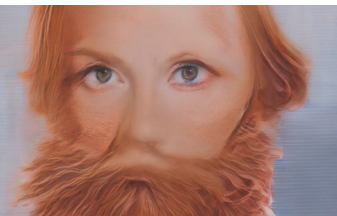
The richness of creative possibilities can risk getting lost in an ocean of options. As for august 2023, more than 15 billion images were created using text-to-image algorithms. Stock photo companies are gradually adding AI-generated images to their catalogs. This is the quantity of photographs that were produced in 150 years, from the first photograph taken in 1826 until 1975.²⁷ From this perspective, AI and bot-generated content, mainly text and images, could soon surpass human-generated content simply because they are easier to produce. Considering that writing is a time-consuming activity for people, a similar fate could occur for textual production, whether literary, journalistic, or academic. Machine-generated content is continuously growing and is also becoming the basis for training future language models. This happens, moreover, in a scenario where, even without AI, produced content in some domains already surpass demand for consumption.

Academic production of papers is one specific example. In popular culture, for instance music, SoundCloud as for mid-2024 has 350 million tracks by 40 million artists, and, according from data from 2023, more than 120.000 tracks were uploaded to streaming services every day. The question therefore arises if we do really need AI songs, if human produced music may have well reached a saturation point, and so many other fields of creative expression is drowning in over-abundance.

[121]



[122]



[121] [122]
Mario
Klingemann,
Makeover
#698
#2606
(2017)

Refik Anadol (2019)

[F] Museum of Modern Art. (n.d.). Refik Anadol: Unsupervised. The Museum of Modern Art. Retrieved December 14, 2025

Machine Hallucinations

Unsupervised is a meditation on technology, creativity, and modern art. Anadol trained a sophisticated machine-learning model to interpret the publicly available data of MoMA's collection. As the model "walks" through its conception of this vast range of works, it reimagines the history of modern art and dreams about what might have been—and what might be to come. In turn, Anadol incorporates site-specific input from the environment of the Museum's Gund Lobby—changes in light, movement, acoustics, and the weather outside—to affect the continuously shifting imagery and sound.

AI is often used to classify, process, and generate realistic representations of the world. In contrast, Unsupervised is visionary: it explores fantasy, hallucination, and irrationality, creating an alternate understanding of art-making itself. The installation is based on works that are encoded on the blockchain, a distributed digital ledger, which stands as a public record of Anadol's art. "I am trying to find ways to connect memories with the future," the artist has said, "and to make the invisible visible. [F]



Amidst the Uncanny

“The Uncanny Landscapes”

Grba, D. (2021). Brittle Opacity: Ambiguities of the Creative AI. Proceedings of the 9th Conference on Computation, Communication, Aesthetics & X (xCoAx 2021), pp. 242- 243.

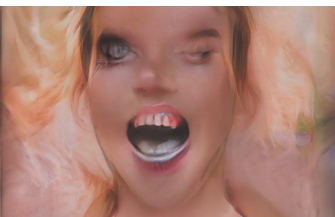
•

The research in artificial intelligence (AI) has been historically inspired by broadly defined concepts of intelligence found in living beings, branching out into various and sometimes conflicted neural models which can be highly-abstract in their relation to cognitive and electrochemical models of the brain (Zheng and Sicker 2013; Sloss and Gustafson 2019). It has been targeting highlevel cognitive functionality which includes the expressions of human intelligence in artistic creativity (Boden 1998; McCorduck 2004). Contemporary AI research centers around a machine learning (ML) paradigm called a neural network, which consists of parameter-processing units (“neurons”), input/ output, and control modules. Such system can be programmed to refine the procedure for solving a particular problem by dynamically modifying parameters based on the referential data. Deep Learning (DL) is a subset of ML methods in which the optimization of network performance and accuracy relies on complex statistical rules applied to multiple layers of neurons (Bishop 2017; Nielsen 2019). The expanding research and implementation of ML have evolved into a range of creative disciplines that engage in the development, application or study of AI, and I use the term creative AI to address the field of these disciplines. The increasing accessibility of DL since 2009 has enabled artists to start exploring the AI systems. Their work contributes in different ways to the creative AI, and shares both the advantages and shortcomings of the field. The existing literature has approached the ambiguities of the creative AI from several viewpoints. Melanie Mitchell in Artificial Intelligence (2019a), as well as Gary Marcus and Ernest Davis in Rebooting AI (2019) provide a conceptual, technological, and social critique of AI focusing on computer science and engineering, media and popular culture.

[123]
[124]
Mario
Klingemann,
Makeover
#3702
#4137
(2017)



[123]



[124]

In *The Ethical Algorithm* (2019), Michael Kearns and Aaron Roth address ethical and social consequences of the conceptual and technical issues in AI algorithm design. In *The Artist in the Machine* (2019), Arthur I. Miller discusses AI art within a study of creativity he had taken in earlier work (1996, 2014). In *AI Art* (2020), Joanna Zylińska provides a multifaceted theoretical discussion of the current AI debate and its reflections in AI-driven visual art. In *Understanding and Creating Art with AI* (2021), Eva Cetini and James She present a comprehensive overview of AI research that takes art as a subject matter, and outline the practical and theoretical aspects of AI art production. With the aim to expand these and other authors’ critical consideration, I address the ambiguities of creative AI from the interrelated perspectives of new media art, mainstream contemporary art and cultural sector, focusing on the poetic and expressive aspects of experimental AI art. This perspective

“THE UNCANNY LANDSCAPES”

Uncanniness is the occasional experience of perceiving a familiar object or event as unsettling, eerie, or taboo (Broad 2020, 36-37), and it can be triggered in close interaction with AI-driven imitations of human physical or behavioral patterns.

Some AI artists approach it implicitly, for example by extracting human-like meaningfulness from machinic textual conversation in Jonas Eltes’ *Lost in Computation* (2017) or by alluding the intimate familiarity of human body in Scott Eaton’s *Entangled II* (2019) (Eaton 2020) which is comparable to earlier video works such as Gina Czarnecki’s *Nascent* and *Spine* (both 2006), and Kurt Hentschläger’s *CLUSTER* (2009-2012) and *HIVE* (2011). AI artworks based on deep fakes, such as Mario Klingemann’s *Alternative Face* (2017) or Libby Heaney’s *Resurrection* (TOTB) (2019), approach uncanniness explicitly by either disrupting or accentuating the formal persuasiveness of statistically rendered human visuals. The artists’ exploration of uncanniness is also related to hybrid artefactual or glitchy aesthetics that can be achieved by emphasizing the abstract visual representations of data in the inner neural layers of DL architectures.

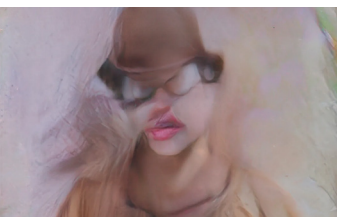
By relying on our pareidolic perception, these visuals play with “humanizing” the opacity of DL processes. This is a popular poetic line in AI art, with examples such as Memo Akten’s *Learning to See* (2017), Mario Klingemann’s *Neural Glitch / Mistaken Identity* (2018b), Weidi Zhang’s *Lavin* (2018), Jukka Hautamäki’s *New Parliament* (2019), and many other.

In *JFK Unsilenced* (*The Greatest Speech Never Made*), a project commissioned by the Times in 2018, Rothco agency took a reminiscent contemplative approach to uncanniness by exploiting the emotional impact of sound, and by referencing the romanticized image of the 35th president of the United States. Based upon the analysis of 831 speeches and interviews, John F. Kennedy’s voice was simulated in a delivery of his address planned for the Dallas Trade Mart on 22 November 1963 (Rothco 2018). At the level of individual words and some short phrases, Kennedy’s voice sounds familiar but overall tone is uneven, so the uncanny effect relies mainly on the context of the speech that young president never had a chance to give. However, even with perfect emulation of Kennedy’s Boston accent, this machinic reincarnation could never come close to matching the eeriness of Kennedy’s televised speech on 22 October 1962. It was contextualized by Cuban missile crisis in which sheer good luck prevented the multilateral confusion, incompetence, ignorance and ultimate insanity of principal human actors from pushing the world into nucleard disaster (Sherwin 2020).

[125] [126]
[127]
Mario
Klingemann,
Makeover
#5723
#5819
#1247
(2017)



[125]



[126]



[127]

Mike Tyka (2019)

[G] Tyka, M. (n.d.). Melissa Yes | Am. Retrieved December 14, 2025

Portraits of Imaginary People

The series, titled “Portraits of Imaginary People” explores the latent space of human faces by training a neural network to imagine and then depict portraits of people who don’t exist. To do so, many thousands of photographs of faces taken from Flickr are fed to a type of machine-learning program called a Generative Adversarial Network (GAN). GANs work by using two neural networks that play an adversarial game: one (the “Generator”) tries to generate increasingly convincing output, while a second (the “Discriminator”) tries to learn to distinguish real photos from the artificially generated ones. At first, both networks are poor at their respective tasks.

But as the Discriminator network starts to learn to predict fake from real, it keeps the Generator on its toes, pushing it to generate harder and more convincing examples. In order to keep up, the Generator gets better and better, and the Discriminator correspondingly has to improve its response. With time, the images generated become increasingly realistic, as both adversaries try to outwit each other. The images you see here are thus a result of the rules and internal correlations the neural networks learned from the training images. [G]



Eating the Future

“The Metabolic Logic of AI Slop”

Crawford, K. Eating the Future: The Metabolic Logic of AI Slop. e-flux, 2025.



An image of a downed US F-8+ jet—wings destroyed, afterburner still glowing—starts to trend. Millions of people watch a video of Tel Aviv in smoking ruins after an Iranian attack. Footage from a US bombing of Tehran takes the place of breaking news from a strike that is yet to happen. Look closer, and you see strange image errors and artifacts. The grass beneath the F-8+ is eerily green and untouched. Things are slightly off. All these images are AI-generated, realistic on first glance, and shared widely on social media and beyond. This is the era of AI slopagenda.

Political leaders are playing a central role in the spread of slop. Supreme Leader Ayatollah Ali Khamenei’s official X account showed an AI image of missile launches rising beneath Quranic verses, viewed over six million times. President Trump frequently shares generated representations of himself as a rock star, a king, and a gold statue in the center of his Gaza resort. In a sense, these images are “classic slop”: hyper-photorealistic, game-engine lighting, and geometrically impossible perspectives. It’s a reconstituted AI mash made from stock photos, commercials, Reddit, and Pinterest. Just add water. Jean Baudrillard once said the simulation comes first and ultimately replaces the real. This barely seems controversial. AI slop is full of revisionist realities, drawing mass attention before the real bombs drop. But in another sense, AI slop isn’t invested in the order of events or even looking like reality. The slop is not the territory: it just smothers it in synthetic goop. It’s flooding the zone with AI shit.

The AI slop aesthetic is more than a technical artifact or cultural curiosity. It’s the consumer-facing product of computational capitalism—created by transforming human culture into a standing reserve of data. Each uncanny image, synthetic influencer, and AI-generated article is part of a much wider transformation. Slop is waste, but it’s also fuel. Slop is the product of a different kind of Krebs cycle. It begins when AI models are trained by devouring billions of images, videos, and texts from the internet, using neural networks with trillions of parameters that learn to identify patterns, relationships, and structures in the data. To digest all this requires staggering amounts of energy. Synthetic outputs are then excreted and then rapidly spread online. Then they are ingested once more, and the cycle continues. This is the latest in a long evolution of industrial media that has been recognized by theorists and artists over decades. We could think of Vilém Flusser’s “technical images” in the Qdets, or Harun Farocki’s “operational images” in the early 1970s. Now we have “metabolic images.” Metabolic images are visual material made by consuming billions of other images to be broken down and: absorbed by AI models via a gargantuan infrastructure that absorbs energy and water, and excretes media, atmospheric carbon, and other pollutants. If we consider all the forms of generative AI, including text, images, code, and video, it is fundamentally metabolic media. It has a burn rate that is astronomical and growing, and is now directly competing with humans for basic resources like land, energy, minerals, and fresh water.

“THE UNCANNY LANDSCAPES”

Karl Marx once described the shift from rural to urban living under capitalism as a disrupted metabolic process. In pre industrial societies, waste easily cycled back into production— human and animal feces from small farms

returned to the earth as fertilizer, maintaining soil fertility and sustaining agriculture. But industrial modernity broke this cycle. As masses of people migrated to cities, the human waste that once nourished fields began accumulating as urban pollution.

People threw excrement into the streets, depriving rural soils of essential nutrients. Soils became depleted, which then required the introduction of artificial fertilizers, exacerbating a cycle of environmental degradation. John Bellamy Foster later called this phenomenon a “metabolic rift”: the systemic disruption of ecological and metabolic processes by capitalist production. Sociologists like Jason W. Moore trace these metabolic rifts back to capitalism’s origins in the sixteenth century, and they resurface in new forms as each phase of industrial development reorganizes the planet. The turn toward massive, data-heavy generative AI is driving a new and dramatic metabolic rift. It’s affecting the environment, patterns of work, supply chains, culture, and visibility itself. And it’s filling information ecosystems with torrents of shit, endless digital detritus which is never properly flushed out. Instead, it is fed back into the system as raw material for the next generation of AI models. This is the basic material cycle of AI’s metabolism: consumption and excretion patterns where AI-generated images have material consequences and consequences for our materials.

Now the widespread use of generative interfaces like ChatGPT, Claude, and Gemini is supplanting web-based information sources. Why bother going to a news site or Wikipedia if Claude can summarize it for you? This diverts users away from the web as it was, toward AI-mediated information platforms. The labor of choosing between lists of different web sites is supplanted by the convenience of being given a single, synthesized answer. This shift is cannibalizing the markets that sustain online content production. Human content creators face dwindling readership, reduced engagement, and disappearing ad revenue. It’s another metabolic rift, severing the relationships between search companies like Google, advertisers, and the broader content economy. In its place, the slop tide will rise. Cursed domains of fake news, psyops imagery, and synthetic influencers designed to generate rage clicks will multiply and capture the paid sponsorships that once belonged to humans. All this poses a problem for AI companies as well. Multiple studies have shown that AI systems degenerate when they are fed on too much of their own outputs—a phenomenon researchers call MAD (Model Autophagy Disease). In other words, AI will eat itself, then gradually collapse into nonsense and noise. It happens slowly at first, then all at once. The researchers compare it to mad cow disease. The industry is scrambling for solutions: engineering higher quality synthetic data for their digital feeding operations and supplementing training

[128]



[129]



[128] [129]
Mario Klingemann, Chicken or Beef #16 #06 (2018)

datasets with content from underpaid human laborers. For humans, a new slop economy is being born. From an infrastructural perspective, the whole cycle is threatening to push past what electrical grids, mines, and water tables can withstand.

“INGEST, DIGEST ARE EXCRETE”

There's an Apple commercial for iPads which serves as an unintentionally perfect metaphor for AI content ingestion. A massive hydraulic press slowly pulverizes musical instruments, video games, books, paint canisters, a typewriter, a camera. Each object collapses under relentless pressure. The ad shows the utter inevitability of the process: no instrument or artwork can tolerate the weight. Centuries of culture and creativity are crushed into a Jne paste for AI training. This is the AI maw in action: the mass harvesting of images, words, videos, all compressed into the latent space of neural networks. Everything we've created or will create, once digitized, is material for the feed lots.

The volume of data required to train generative AI systems is rapidly approaching the limits of what exists on the open web. Some researchers say this will happen by)*) k. Elon Musk argues that we have already reached “peak data.” The AI industry guards the exact details, but reports suggest earlier GPT-8 models were trained on more than R+ terabytes of compressed internet content, at least 8.) trillion words scraped from websites, books, articles, and social media posts created by millions of human authors. The latest public datasets like Public Domain Q)M contain more than Q).R billion images with text captions. AI companies now acquire entire social media platforms and negotiate exclusive deals with libraries and academic journal publishers, but it may not be enough to meet the growing appetites of AI models. Digesting all that media requires the energy of an industrialized country. Estimates of how much electricity will be needed for generative AI is now routinely expressed in the unit of a nation state. The International Energy Agency says that by)*8*, AI will consume as much electricity as Japan today. Bloomberg predicts that AI will need as much power as India by the same date. But AI's demands are growing faster than any individual country.

Each generative AI query burns at least ten times more power than traditional web searches. Text-to-image generation is even more energy intensive, while video generation is exponentially greater again. This escalating industrial demand follows the recursive logic of AI development itself: larger models demand more data, which enables even larger models, producing higher resolution outputs, requiring more computing power, consuming

more energy—and repeat. Consider the latest AI hyper-scale data center being built in Abilene, Texas for OpenAI—code-named “Project Ludicrous.” When completed, it will span more than seventeen football Jelds. Elon Musk's xAI facility in South Memphis, called Colossus, which was built in)*)R, has already hit the limit of the local electrical grid. The company installed methane gas generators that emit toxic nitrogen dioxide and formaldehyde into the air. The predominantly Black community who live in the region is already experiencing high levels of asthma and respiratory disease. And this is merely the beginning. President Trump's Stargate Project is planning at least twenty giant data centers across the US. The industry calls them AI factories.

Then there's water. Like digestive `uid, hyperscale data centers require continuous cooling systems that consume millions of gallons daily. With billions worldwide using generative AI, aquifers of drinking water are being evaporated to cool down AI chips. This represents a vast atmospheric redistribution of hydrological resources taken away from source ecosystems and dispersed as vapor. Finally, there is a new critical minerals “supercycle” driven by burgeoning AI infrastructure, which is accelerating environmental damage. Rare earth elements essential for GPUs, predominantly mined and processed in China, leave behind lakes of radioactive waste. Lithium for data center backup systems comes from salt `ats in Chile and Bolivia, disrupting local water systems and threatening Indigenous land rights. The cobalt for batteries relies on mining operations in the Democratic Republic of Congo that exploit child labor and operate with minimal environmental oversight. The expansion of computational capital reproduces the core periphery dynamics of earlier empires—extracting from the Global South to enrich the North—while excavating billions of years in geological time to increase the speed of algorithmic convenience by milliseconds. After feeding at this planetary scale, the industrial beast now releases a tide of slop.

“SLOP ECONOMICS”

How are the humans who live downstream of AI models adapting? Do we swim with or against the rivers of synthetic shit? If the metabolic rift decimates our ecosystems—including the information environment—whatever will become of the in`uencers? They're transforming into coprophages: digital dung-eaters that consume and monetize slop.

This metamorphosis is already underway. AI-powered personas like Lil Miquela achieve engagement rates far beyond human in`uencers. Their creators systematically

[130] [131]
Mario
Klingemann,
Chicken
or Beef
#29
#25
(2018)

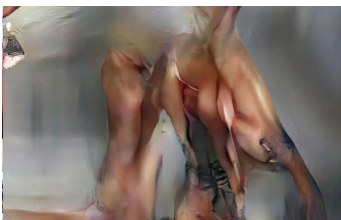


[130]

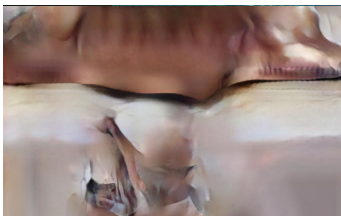


[131]

[132]



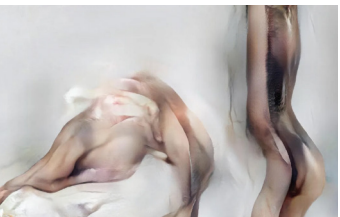
[133]



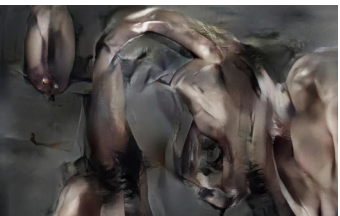
[134]



[132]
[133][134]
Mario
Klingemann,
Chicken
or Beef
#38
#36
#17
(2018)



[135]



[136]

[135] [136]
Mario
Klingemann,
Chicken
or Beef
#32
#24
(2018)

A/B test visual aesthetics, personality traits, and content forms to maximize dwell time. These characters appear increasingly indistinguishable from living humans while being ruthlessly optimized for audience response and commercial electiveness. Content creation teams deploy AI tools to generate hundreds of potential posts, images, and videos, then use engagement analytics to identify the most addictive combinations. Instagram accounts are dedicated to AI-generated “rage bait” in`uencers engineered to provoke, olend, and harvest outrage. These synthetic personalities are continuously evolving based on performance data, with unsuccessful traits eliminated and successful characteristics amplified before being spawned into new accounts.

This industrialization of viral content is a well-established practice under platform capitalism. Instagram doesn’t reveal how much it pays creators once their content spreads widely, but recent reporting suggests it’s roughly \$Q)* per million views. The system proves so indiscriminate that AI-generated images now earn creators hundreds of dollars per viral hit, transforming Meta’s \$) billion creator fund into an inadvertent subsidy for synthetic content. The platform has thus engineered the perfect economic incentive structure for slop: rewarding whoever can most eVciently capture human attention, regardless whether it’s ASMR fruit-cutting videos, bunnies jumping on trampolines, or war propaganda. Meta calls this “supporting creators,” but with more people monetizing generative AI, it’s sponsoring slop factories. The global geography of synthetic content production reveals the international division of labor underlying these new economic relationships. Content farms in regions with lower labor costs deploy AI tools to generate thousands of synthetic social media posts, news articles, and product reviews targeting higher-value advertising markets in wealthy nations. It’s a well-oiled form of synthetic arbitrage: the extraction of ad revenue from developed economies through mass production of AI slop in low-income regions. Will AI slop farming ultimately exhaust the soil of online spaces? Will people `ee the slop, tolerate it, or even welcome it? Will we ultimately come to love slop?

“WITHER SLOP?”

AI slop is not an aberration but an inevitable feature of how generative media works. Yet the current arrangements are both systemically and inherently unstable. The metabolic rift between humans and AI threatens multiple forms of failure: model collapse, ecological collapse, and cognitive collapse. The current approach to AI cannot sustain itself indefinitely. The question becomes not whether current AI slop economies will destroy themselves, but when. This instability creates space for what we might call “post

synthetic” cultural organization: forms of culture operating under dilerent metabolic logics, whether by design or necessity. Such alternatives would demand the fundamental transformation of existing technical and economic arrangements—alternative energy systems, cooperative ownership models, and cultural protocols that resist algorithmic ingestion. This requires not a nostalgic return to pre-digital cultural forms, but practices adequate to our ecological and social moment.

The question that remains is whether we will choose to embrace slop or cultivate desire for something else entirely: cultural forms moving beyond synthetic reproduction, oriented toward `ourishing rather than extraction; toward ecological regeneration rather than a hyper-metabolic state of exhaustion and breakdown.



Mario Klingemann, "Deposition #1"
- From Deposition 2018

On Aesthetics
Where We Have Been and Where We Are Going

DESIGN AND CONCEPT
Iris Buljević

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